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RESEARCH ARTICLE

CHARD: Characteristic-Aware Hierarchical Detection for Bangladesh Road Scenes

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ABSTRACT Traffic in South Asia is dense, weakly structured, and includes vehicle types such as auto-rickshaws, pedal-powered three-wheelers, and cart vehicles that are absent from standard benchmarks such as COCO, BDD100K, and Waymo. Detection is further challenged by severe class imbalance, where the locally important vehicle classes are also the rarest. We propose Characteristic-Aware Hierarchical Detection (CHARD), which uses vehicle characteristics, wheel count, size, and propulsion as auxiliary supervision during training. This allows rare classes to learn from shared attributes instead of relying only on limited class-specific examples. CHARD is implemented as an end-to-end hierarchical attribute-to-class head for RT-DETR and as a lightweight auxiliary attribute branch for YOLO that adds no parameters or inference latency. We evaluate 72 independently trained models (eight variants, three random seeds) on three Bangladeshi road-vehicle datasets: BadODD, Poribohon-BD, and Sorokh-Poth. On BadODD, CHARD with a YOLOv11-l backbone achieves the highest seed-averaged test mAP@50:95 of 44.3%, outperforming its no-attribute counterpart by 2.4 points ($p = 0.003$) and the standard YOLOv11-l baseline by 1.8 points. The improvement is concentrated on tail classes, where AP increases from 28.0% to 31.9%, while head and medium-frequency classes remain unchanged. CHARD also matches or exceeds the baseline across all frequency groups on both validation and test sets without increasing inference cost. The benefit is conditional rather than universal. Auxiliary supervision is statistically neutral on the lower-capacity YOLOv8-l backbone and on the two supplementary datasets with weaker class imbalance. These results suggest that characteristic-aware supervision is most effective when sufficient model capacity and pronounced long-tail distributions allow shared attributes to improve rare-class learning.

INDEX TERMS Autonomous driving, Bangladesh, empirical analysis, hierarchical detection, long-tail classification, negative result, object detection.

I. INTRODUCTION

Autonomous driving has progressed rapidly over the past decade, with large-scale benchmark datasets such as KITTI [1], Cityscapes [2], BDD100K [3], and the Waymo Open Dataset [4] catalyzing advances in perception. These resources, however, predominantly capture road environments in North America, Europe, and East Asia. The traffic ecosystems of the Indian subcontinent are densely populated, weakly structured, and characterized by an unusually heterogeneous mix of vehicles, and remain comparatively under-represented. Achieving robust autonomous navigation in these contexts requires datasets and models that explicitly address the distinctive composition of South Asian roads, including locally manufactured vehicles such as auto-rickshaws, pedal-powered three-wheelers, and animal- or human-propelled cart vehicles that have no direct analog in widely used benchmarks [5].

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The recently released Bangladeshi Autonomous Driving Object Detection (BadODD) dataset [6] addresses this gap with 10,032 smartphone-captured images collected across nine districts, comprising 80,036 annotated instances spanning 13 classes that reflect the actual traffic composition of Bangladesh. This composition exposes a challenge that is more than academic: the locally specific vehicle types that matter most for safe navigation (e.g., wheelchairs and cart vehicles sharing the road with motorized traffic) are also the rarest, with an instance ratio of 870:1 between the most and least frequent classes. A distinguishing feature of BadODD is its accompanying characteristic-based taxonomy, in which each class is described by three attributes: wheel count, size, and propulsion mechanism. This design choice was motivated by the difficulty of assigning consistent labels to vehicles that lack globally acknowledged names, and it raises a natural algorithmic question: can this taxonomy serve as a useful supervisory signal that lets these rare, safety-critical classes borrow statistical strength from their more common attribute-sharing neighbors?

Despite the practical importance of BadODD, no systematic algorithmic study has been conducted on it. The original dataset paper [6] reports only preliminary baselines using YOLOv5 and YOLOv8 with limited training, and provides no analysis of geographic or per-class behavior. Two questions follow directly: *which modern detector architectures perform best on BadODD?* and *can the characteristic taxonomy be exploited algorithmically beyond its role in label assignment?* A third question, less obviously motivated but no less important, concerns the dataset's geographic structure: BadODD spans nine districts that differ substantially in road type, traffic density, and illumination conditions. Whether aggregate performance metrics adequately capture model behavior across these conditions is an open question with direct implications for deployment.

In this work, we conduct a comprehensive empirical study addressing these three questions. Our analysis is organized around the following research questions:

- **RQ1.** How do modern object detection architectures compare on BadODD?
- **RQ2.** Does the BadODD characteristic taxonomy provide useful auxiliary supervision when integrated into detection training, either as a hierarchical head replacement or as a parallel auxiliary task?
- **RQ3.** How does geographic variation across the nine districts of BadODD affect detection quality, and how does this variation compare to inter-model variation?

To answer these questions, we benchmark four recent detectors: YOLOv8-l [7], YOLOv10-l [8], YOLOv11-l [9], and RT-DETR-l [10] on the standard BadODD splits. We propose CHARD (Characteristic-Aware Hierarchical Detection), a framework that exploits the BadODD taxonomy through hierarchical attribute-then-class prediction. CHARD is evaluated in two integration strategies: an end-to-end variant built on RT-DETR with a custom hierarchical head and training pipeline, and a lightweight auxiliary-task variant that augments a YOLO detector with parallel attribute supervision at task-aligned anchors and is discarded before inference, so the deployed model incurs no additional parameters or latency. We instantiate the auxiliary variant on two backbones of differing capacity (YOLOv8-l and the stronger YOLOv11-l), which turns out to be decisive: the same characteristic supervision is neutral on YOLOv8-l but yields a significant improvement on YOLOv11-l. To verify that this conclusion reflects a genuine architectural effect rather than the noise of a single training run on a single dataset, we repeat every configuration under three random seeds and replicate the full comparison on two further Bangladeshi road-vehicle datasets, Poribohon-BD and Sorokh-Poth, for a total of 72 independently trained models. We further introduce a per-district stratified evaluation protocol that complements the dataset's standard random split and reveals geographic difficulty patterns that aggregate metrics conceal.

Our contributions are as follows:

1) We propose **Characteristic-Aware Hierarchical Detection (CHARD)**, a framework that incorporates vehicle characteristics (wheel count, size, and propulsion) as auxiliary supervision for object detection. We implement CHARD as both an end-to-end hierarchical head for RT-DETR and a lightweight training-only auxiliary branch for YOLO, introducing no additional inference cost.
2) We conduct the first comprehensive benchmark of modern object detectors on Bangladeshi road scenes using BadODD, complemented by experiments on Poribohon-BD and Sorokh-Poth. Our evaluation spans 72 independently trained models across multiple architectures, datasets, and random seeds.
3) We show that characteristic-aware supervision is *conditional* rather than universally beneficial. It significantly improves long-tail detection on the YOLOv11-l backbone, while remaining statistically neutral on YOLOv8-l and on datasets with weaker class imbalance, demonstrating that its effectiveness depends on both model capacity and data distribution.
4) We introduce a geographically stratified evaluation protocol and show that detection difficulty varies by nearly 2× across districts, substantially exceeding inter-model differences. This highlights the importance of reporting geographic performance alongside aggregate detection metrics.

The remainder of the paper is organized as follows. Section II reviews relevant work on autonomous driving datasets, object detection architectures, and hierarchical and auxiliary supervision. Section III characterizes BadODD along the dimensions relevant to our study, including class distribution, geographic distribution, and the characteristic taxonomy, and introduces the two supplementary datasets, Poribohon-BD and Sorokh-Poth, used for cross-dataset replication. Section IV describes the evaluated architectures and the CHARD framework across its integration strategies and backbones. Section V details the experimental setup. Section VI presents the main results across all three evaluation protocols. Section VII discusses the principal findings, including the backbone-dependent behavior of CHARD and the geographic difficulty analysis. Section VIII documents what is fixed and what is released to make the study reproducible. Section IX acknowledges limitations and outlines avenues for future work, and Section X concludes.

II. RELATED WORK

This section reviews prior research relevant to our study, including autonomous driving datasets, modern object detection architectures, and approaches that leverage hierarchical or auxiliary supervision. By examining the strengths, limitations, and findings of existing literature, we establish the research context for BadODD and identify the gaps that motivate our empirical evaluation and the proposed CHARD framework.

A. AUTONOMOUS DRIVING DETECTION DATASETS

The development of object detection for autonomous driving has been propelled by progressively larger and more diverse datasets. KITTI [1] introduced LiDAR-camera multimodal annotations on German highways and urban streets, and remains a standard 3D benchmark. Cityscapes [2] expanded scene-level coverage to 50 European cities with fine-grained semantic annotations. BDD100K [3] substantially increased scale and weather/time-of-day diversity across the United States, providing 100,000 video sequences with annotations for ten common object categories. The Waymo Open Dataset [4] introduced production-grade sensor suites combining LiDAR and multiple cameras, with high-quality 3D annotations for four object classes (vehicles, pedestrians, cyclists, and signs) captured in suburban and highway driving in the western United States. While not autonomous-driving-specific, COCO [11] remains the de facto pretraining and benchmarking dataset for general object detection.

These resources share a common geographic limitation: they capture road environments in North America, Europe, or East Asia, where traffic is comparatively structured and vehicle types are relatively homogeneous. The Indian Driving Dataset (IDD) [5] was the first large-scale effort to address South Asian conditions, with 10,004 images collected in India and annotations for 34 classes that include culturally specific vehicles. IDD established that detectors trained on Western datasets transfer poorly to unstructured South Asian traffic, and motivated subsequent regional benchmarks. This line of work has continued to expand: DriveIndia [12] contributes 66,986 images over 24 traffic-relevant classes collected across urban, rural, and highway routes under varied weather and illumination, confirming that regionally-specific benchmarks remain an active and necessary research direction. BadODD [6] extends this line of work to Bangladesh with roughly 10,000 smartphone-captured images across nine districts and 13 classes selected to reflect the local vehicle ecosystem. BadODD is unique in two respects: its exclusive use of smartphone (rather than automotive-grade) cameras, and its accompanying characteristic-based taxonomy describing each class by wheel count, size, and propulsion mechanism. Bangladeshi road scenes have attracted growing attention in parallel: the Bangladeshi Native Vehicle Dataset [13] annotates 81,542 instances over 17 native vehicle classes collected across 17 districts, and Hossain et al. [14] compare YOLO generations for real-time vehicle detection in Bangladeshi urban environments. These efforts are complementary to ours in that they enlarge and benchmark the data available for the region; none, however, exploits a characteristic taxonomy as a supervisory signal, which is the question we pursue here.

B. OBJECT DETECTION ARCHITECTURES

Modern object detectors fall broadly into anchor-based, anchor-free, and query-based paradigms. The YOLO family has dominated the real-time detection landscape through a sequence of architectural refinements, comprehensively surveyed by Terven and Córdova-Esparza [15]. YOLOv8 [7] adopted an anchor-free decoupled head with distribution focal loss and task-aligned assignment, achieving a strong speed–accuracy balance. YOLOv10 [8] removed non-maximum suppression through consistent dual-label assignment, enabling true end-to-end inference. YOLOv11 [9] introduced the C3k2 block and SPPF refinements, further improving accuracy at comparable inference cost. The family continues to evolve toward attention-based designs: YOLOv12 [16] replaces much of the convolutional stack with an area-attention mechanism and reports accuracy gains over YOLOv10 and YOLOv11 at comparable latency. We evaluate through YOLOv11-l, the most recent version with a mature and widely deployed training implementation at the time of our experiments, and note that extending our characteristic-supervision study to attention-centric backbones is a natural direction for future work.

The DETR family [17] reformulated detection as a set prediction problem using bipartite matching and transformer decoders, eliminating hand-crafted components such as anchors and NMS. Deformable DETR [18] addressed the slow convergence of the original DETR through multi-scale deformable attention. DINO [19] introduced contrastive denoising training, mixed query selection, and look-forward-twice box refinement to achieve state-of-the-art results. RT-DETR [10] adapted the DETR paradigm for real-time inference by combining an efficient hybrid encoder with IoU-aware query selection, becoming the first DETR variant to surpass YOLO at comparable latencies. RT-DETRv2 [20] subsequently refined this design with scale-adaptive sampling in the deformable attention, a discrete sampling operator that removes DETR-specific deployment constraints, and dynamic data augmentation, improving accuracy without additional inference cost. The real-time query-based line has advanced rapidly since: D-FINE [21] recasts box regression as fine-grained distribution refinement with self-distillation of localization, and DEIM [22] accelerates convergence through dense one-to-one matching and a matchability-aware loss, together substantially narrowing the training-cost gap that has historically disadvantaged DETR-style detectors relative to YOLO. Our study evaluates the leading representatives of both paradigms (YOLOv8/10/11 and RT-DETR) and uses RT-DETR as the base architecture for our custom CHARD variant; the brittleness we observe for the RT-DETR family under dataset shift (Section VI-F) suggests that these more recent stabilizing refinements are a promising avenue for improving the query-based branch of our framework.

C. LONG-TAIL AND HIERARCHICAL CLASSIFICATION

Object detection datasets typically exhibit long-tailed class distributions, with BadODD being an extreme case (Person accounts for 38% of instances, Construction Vehicle for about 0.05%). The class-balanced loss of Cui et al. [23] reweights cross-entropy by the effective number of samples per class. Focal loss [24] addresses imbalance through a modulating factor that down-weights confident predictions, and remains the standard formulation for sigmoid-based detection heads. Equalization losses [25] and Seesaw loss [26] explicitly suppress gradient interference from frequent classes onto rare ones, achieving strong results on the LVIS long-tail benchmark [27]. Zhang et al. [28] survey this field comprehensively, organizing methods into class re-balancing, information augmentation, and module improvement. Work in this area remains active: Ma et al. [29] argue that raw instance counts are a poor proxy for learning difficulty and instead reshape decision boundaries using a per-category information-amount measure, a perspective directly relevant to BadODD, where our tail classes differ by orders of magnitude in frequency yet share attribute structure. Two further recent directions are worth contrasting with ours. SimLTD [30] attacks the tail through a staged supervised and semi-supervised curriculum that pre-trains on head classes before transferring to scarce tail classes, and FRACAL [31] operates entirely post hoc, adjusting logits at inference using the fractal dimension of each class's spatial distribution. Both intervene on the data regime or the decision rule; CHARD instead injects structure through the label space itself during training, which is why its benefit is contingent on the backbone having capacity to absorb the auxiliary signal. Hierarchical classification approaches [32], [33] have leveraged taxonomic structure to share representations across related classes, with empirical evidence that hierarchical losses help when the taxonomy reflects genuine semantic relationships unavailable in the flat class labels.

D. AUXILIARY TASK LEARNING IN DETECTION

Auxiliary supervision augments the primary detection objective with related prediction tasks. Mask R-CNN [34] demonstrated that segmentation supervision improves detection accuracy. Subsequent work has explored auxiliary tasks including depth estimation [35], keypoint prediction, and attribute classification [36]. The principal hypothesis is that auxiliary tasks regularize shared representations and encourage features that capture richer semantic structure. However, the benefits are not universal: Standley et al. [37] showed that task combinations can interfere when objectives compete for representational capacity, and subsequent work [38] has emphasized careful task selection and loss balancing. This failure mode, termed *negative transfer*, has become a central concern in recent auxiliary-task learning research. Jiang et al. [39] characterize it directly and mitigate it by periodically forking the model into branches, searching task weights against target validation error, and merging to filter out detrimental parameter updates. Their framing is instructive for our results: whether an auxiliary task helps is not a fixed property of the task pair but depends on how much capacity the shared representation can allocate without degrading the primary objective. Our CHARD framework follows this auxiliary-task tradition by introducing characteristic prediction (wheel, size, propulsion) as a parallel supervisory signal, and our finding that the identical auxiliary task is neutral on one backbone yet beneficial on a higher-capacity one (Section VII-B) provides empirical support for exactly this capacity-dependent view.

III. DATASET CHARACTERIZATION

This section presents the key characteristics of the BadODD dataset that are most relevant to our analysis. We examine its class distribution, geographic coverage, and characteristic-based taxonomy, highlighting the unique properties of Bangladeshi road scenes and the challenges they introduce for object detection models. These dataset insights provide the foundation for the experimental design and evaluation presented in later sections.

BadODD [6] consists of 10,032 images partitioned into training (5,967), validation (2,032), and testing (2,033) splits in a 60:20:20 ratio. The dataset contains 80,036 annotated bounding boxes across 13 vehicle and pedestrian classes. All images were collected using smartphone cameras across nine districts of Bangladesh under varying lighting conditions (day and night) and road types (urban, rural, expressway, and village roads), ensuring representativeness of real driving scenarios that an autonomous vehicle would encounter in this context. All counts reported in this paper are computed directly from the released distribution we obtained and used for every experiment; they differ marginally from the summary figures quoted in the original dataset paper, and we report the measured values so that every statistic here is reproducible from the released splits.

A. CLASS DISTRIBUTION

The class distribution is extremely long-tailed. Table 1 reports the instance count and percentage of each class across the three splits. Person and auto-rickshaw together account for over 60% of all instances, while construction vehicle, train, and wheelchair each represent less than 0.1% of the training data. The ratio between the most and least frequent classes reaches 870:1, placing BadODD among the most imbalanced detection datasets relative to its class count.

B. FREQUENCY-BASED CLASS GROUPINGS

To facilitate the long-tail analysis presented later in this work, we partition classes into three frequency groups based on their share of training instances. Following the convention used in long-tailed recognition benchmarks [26], [27], we use a head/medium/tail split with thresholds at 10% and 3% of total instances. The resulting groups are listed in Table 2.

TABLE 1. Per-class instance distribution across the BadODD splits. Percentages are computed within each split.

Class	Train	%	Val	%	Test	%
Person	18,268	38.45	6,423	39.24	6,325	39.16
Auto-rickshaw	10,642	22.40	3,661	22.37	3,517	21.77
Three-wheeler	5,732	12.06	1,948	11.90	1,984	12.28
Car	3,786	7.97	1,314	8.03	1,236	7.65
Motorbike	3,771	7.94	1,235	7.54	1,266	7.84
Truck	2,304	4.85	777	4.75	782	4.84
Bus	1,885	3.97	591	3.61	620	3.84
Bicycle	678	1.43	261	1.59	256	1.58
Priority vehicle	230	0.48	73	0.45	66	0.41
Cart vehicle	144	0.30	48	0.29	46	0.28
Wheelchair	29	0.06	27	0.16	28	0.17
Construction vehicle	24	0.05	6	0.04	11	0.07
Train	21	0.04	5	0.03	16	0.10
Total	**47,514**	**100**	**16,369**	**100**	**16,153**	**100**

TABLE 2. Frequency-based class groupings used for long-tail analysis.

Group	Classes
Head	Person, Auto-rickshaw, Three-wheeler
Medium	Car, Motorbike, Truck, Bus
Tail	Bicycle, Priority Vehicle, Cart Vehicle, Wheelchair, Construction Vehicle, Train

C. GEOGRAPHIC DISTRIBUTION

The nine districts represented in BadODD differ substantially in urbanization, road infrastructure, and traffic composition. Table 3 reports the fraction of training images collected from each district. The largest district (Dhaka) contributes 12× more images than the smallest (Rajshahi), reflecting a natural imbalance that mirrors population density and accessibility. This distribution is one factor we revisit in our per-district evaluation (Section VI); district-level sample size is only weakly associated with detection difficulty, however; the smallest district, Rajshahi (96 images), is among the easier ones; so we treat scene and imaging characteristics, rather than image count, as the primary drivers of the geographic gap.

D. VISUAL DIVERSITY

Figure 1 shows representative annotated images from all three datasets used in this study. The top three rows sample one image from each of the nine BadODD districts. The visual diversity spans daytime urban scenes (Dhaka, Chittagong) with high traffic density and visible auto-rickshaws and three-wheelers; expressway scenes (Maowa) with sparser vehicle distributions; rural settings (Sherpur, Sirajganj) with mixed motorized and non-motorized traffic; and night scenes with markedly different illumination characteristics. The smartphone-camera capture introduces variation in resolution, motion blur, rolling shutter artifacts, and color rendition that automotive-grade datasets do not exhibit, and even the image geometry varies by district: the Sirajganj subset is captured exclusively in portrait orientation, which is why that panel is padded rather than cropped.

TABLE 3. District-level distribution of training images in BadODD.

District	Images	%
Dhaka	1,158	19.41
Mymensingh	1,028	17.22
Sylhet	851	14.27
Khulna	807	13.52
Chittagong	677	11.35
Maowa	637	10.68
Sherpur	438	7.34
Sirajganj	275	4.61
Rajshahi	96	1.61
Total	5,967	100.00

The bottom two rows show the supplementary datasets after conversion to the shared taxonomy (Section III-G), with class names printed on each box so that the mapping can be inspected directly. They make the differences quantified in Table 4 visible: Poribohon-BD contributes moderately populated street scenes, while Sorokh-Poth is typically centered on a single annotated vehicle. The panels also illustrate the locally-named categories that motivate the mapping, including a leguna annotated as `truck`, a van (a human-pulled flatbed cargo cart) as `cart_vehicle`, and a cng as `auto_rickshaw`.

E. BOX SIZE DISTRIBUTION

Beyond class imbalance, BadODD also exhibits substantial variation in bounding box size, which influences detection difficulty. Figure 2 (left) presents the distribution of normalized box areas across all training instances; the distribution is heavily skewed toward small objects, with the median normalized box area below 0.5% of the image. Figure 2 (right) shows the average box area per class, revealing that head classes (person, auto-rickshaw, three-wheeler) include many small instances while tail classes such as bus and train, when present, occupy comparatively larger image regions.

F. CHARACTERISTIC TAXONOMY

A distinguishing feature of BadODD is its accompanying characteristic-based taxonomy, in which each of the 13 classes is described by three attributes: wheel count {2, 3, 4, other}, size {small, medium, big}, and propulsion {paddle, fossil, electric, human, animal}, where paddle is the

FIGURE 1. Representative annotated images from the three datasets. Rows 1–3: one training image from each of the nine BadODD districts, spanning daytime, night, urban, rural, and expressway conditions. Row 4: Poribohon-BD. Row 5: Sorokh-Poth. All boxes are ground-truth annotations after mapping to the common 13-class taxonomy; class names are printed for the two supplementary datasets, whose sparser scenes make the mapping legible, and omitted for the denser BadODD panels. Panels are padded, never cropped, to a common aspect ratio, so every annotated object remains visible.

dataset's term for pedal-powered. This taxonomy was introduced to address the ambiguity of vehicle naming conventions in the region and to support future extension to newly introduced vehicle types. Importantly for our study, the taxonomy provides an additional supervisory signal beyond the flat class labels: for example, the rare wheelchair class

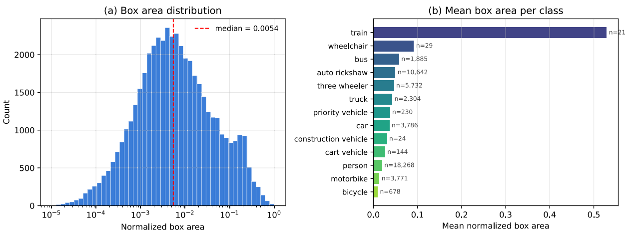

FIGURE 2. Bounding box statistics for BadODD training instances. Left: Histogram of normalized box areas (log scale). Right: Mean normalized box area per class. The dataset is dominated by small objects, with median box area below 0.5% of the image.

shares its (2-wheel, small, human) attribute combination partially with bicycle and pedestrian categories, suggesting that attribute-level supervision could in principle improve generalization on tail classes. Section IV formalizes this idea, and Sections VI and VII report empirical findings on whether the taxonomy actually delivers this benefit.

G. SUPPLEMENTARY DATASETS: PORIBOHON-BD AND SOROKH-POTH

Conclusions drawn from a single dataset risk reflecting the idiosyncrasies of that dataset rather than properties of the architectures under test. We therefore replicate our entire model comparison on two further Bangladeshi road-vehicle datasets, chosen because they cover the same vehicle ecosystem while differing substantially in scene composition and class balance.

Poribohon-BD [40] is a Bangladeshi local-vehicle image dataset collected from smartphone cameras and social media, annotated for classification and detection of native vehicle types. We use a publicly redistributed detection-annotated release comprising 5,205 images and 13,346 boxes, an average of 2.56 objects per image.

Sorokh-Poth [41] is a deliberately class-balanced Bangladeshi road-vehicle dataset distributed as per-class image folders with Pascal-VOC annotations. It contributes 4,450 images and 4,462 boxes. Its defining property is that images are typically centered on a single annotated vehicle (1.00 objects per image), making it a far sparser and visually simpler benchmark than BadODD.

Both datasets use native vehicle vocabularies that differ from BadODD's. We map them onto the common 13-class BadODD taxonomy with a released conversion script, resolving locally-named categories by visual inspection: `cng` and `easybike` (motorized and electric three-wheeled taxis) map to `auto_rickshaw`; the pedal-powered `rickshaw` maps to `three_wheeler`; `leguna` (a converted pickup-chassis shared transport) maps to `truck`; and `van`, which in Bangladeshi usage denotes a human-pulled flatbed cargo cart rather than a passenger van, maps to `cart_vehicle`. Sorokh-Poth is distributed without a predefined partition, so we impose a deterministic stratified 70:20:10 train/validation/test split, sampling within each source class so that all splits remain balanced.

Table 4 contrasts the three datasets. Two differences are material to the interpretation of our cross-dataset results. First, object density differs by nearly an order of magnitude (7.98 objects per image in BadODD versus 2.56 and 1.00), so the supplementary datasets pose a substantially easier detection problem and yield correspondingly higher absolute mAP; they test the *ranking* of architectures under distribution shift rather than providing comparable difficulty. Second, and more consequentially, both supplementary datasets cover only eight of the thirteen BadODD classes: `person`, `priority_vehicle`, `construction_vehicle`, `train`, and `wheelchair` are entirely absent. Their class distributions are therefore far flatter than BadODD's 870:1 ratio, with Sorokh-Poth in particular spanning only 8.4%–19.3% per class. Neither dataset contains the extreme long tail that motivates the CHARD framework, a point we return to when interpreting why characteristic supervision helps on BadODD but is neutral elsewhere (Section VII-B).

TABLE 4. The three datasets used in this study, after mapping to the common 13-class BadODD taxonomy. Imbalance is the instance ratio between the most and least frequent present class.

	BadODD	Poribohon-BD	Sorokh-Poth
Images	10,032	5,205	4,450
Annotated boxes	80,036	13,346	4,462
Objects / image	7.98	2.56	1.00
Classes present	13	8	8
Imbalance ratio	870:1	26:1	2.3:1
Most frequent	person 38%	auto-rick. 20%	truck 19%
Districts labeled	9	–	–

H. SUMMARY

BadODD presents three concurrent challenges that distinguish it from established autonomous driving benchmarks: an 870:1 class imbalance ratio across 13 classes, substantial geographic diversity across nine districts with markedly different traffic and lighting conditions, and exclusive smartphone-camera capture rather than automotive-grade sensor data. The characteristic-based taxonomy provides additional structure that may or may not translate into useful supervision; testing this empirically is one of the principal contributions of this work. Poribohon-BD and Sorokh-Poth complement BadODD as replication targets: they share its vehicle ecosystem but present sparser scenes and much flatter class distributions, which lets us test whether our architectural conclusions survive a change of dataset.

IV. METHODS

This section describes the methodology adopted in this study, including the selected object detection models, the proposed Characteristic-Aware Hierarchical Detection (CHARD) framework, and the overall experimental pipeline.

We outline how the models are trained and evaluated on the BadODD dataset to investigate the impact of characteristic-aware supervision on object detection performance in complex driving environments.

A. EVALUATED ARCHITECTURES

We evaluate four modern detectors on BadODD, selected to span both single-stage anchor-free designs (the YOLO family) and query-based set prediction (RT-DETR). All models are used at their large ("-l") configuration with input resolution 640 × 640 for a fair compute comparison.

1) YOLOv8-L

YOLOv8 [7] adopts an anchor-free decoupled detection head with separate convolutional branches for classification and box regression. Box localization uses Distribution Focal Loss [42], and target assignment is performed by the Task-Aligned Assigner [43], which jointly considers classification confidence and localization quality when matching predictions to ground truth. The large variant contains approximately 43.7M parameters with 165.4 GFLOPs.

2) YOLOv10-L

YOLOv10 [8] removes non-maximum suppression at inference time through a consistent dual-label assignment scheme that combines one-to-many supervision during training with one-to-one assignment for end-to-end inference. The architecture introduces holistic efficiency-accuracy refinements, including spatial-channel decoupled downsampling and a rank-guided block design. The large variant has approximately 25.7M parameters and is reported by the authors to match YOLOv9 accuracy at lower latency.

3) YOLOv11-L

YOLOv11 [9] introduces the C3k2 block, an evolution of the C2f module used in YOLOv8, providing improved feature aggregation at comparable computational cost. The detection head retains the anchor-free decoupled design from YOLOv8 while incorporating spatial attention mechanisms in the neck. The large variant balances accuracy and efficiency improvements and serves as our primary reference baseline in our experiments (one of the two co-leading baselines, as the multi-seed study confirms).

4) RT-DETR-l

RT-DETR [10] is the first DETR variant to surpass YOLO models in real-time inference. It combines an efficient hybrid encoder that decouples intra-scale attention from cross-scale fusion with IoU-aware query selection that initializes object queries from high-quality encoder predictions. The decoder employs standard transformer cross-attention with auxiliary supervision on every decoder layer. The Ultralytics RT-DETR-l we benchmark uses an HGNetv2 backbone with approximately 32M parameters and 103.5 GFLOPs. Our custom CHARD-RT-DETR variant (Section IV-B2) is instead built on the HuggingFace ResNet-50-vd checkpoint (`PekingU/rtdetr_r50vd_coco_o365`, roughly 42M parameters), a different backbone; we therefore treat CHARD-RT-DETR as a standalone implementation rather than a matched variant of the RT-DETR-l baseline.

B. CHARD: CHARACTERISTIC-AWARE HIERARCHICAL DETECTION

The BadODD characteristic taxonomy (Section III) assigns to each class $c \in \{1, \ldots, C\}$ a tuple of three attributes $a_c = (w_c, s_c, p_c)$ describing wheel count, size, and propulsion. While the class labels themselves exhibit extreme imbalance (an 870:1 ratio between Person and Train), the attribute distributions are substantially more uniform: the rare Wheelchair class shares its (*2-wheel*, *small*, *human*) signature partially with Bicycle (678 training instances) and Person (18,268 instances). This observation motivates a hierarchical prediction scheme in which attributes are predicted first, and the class is then conditioned on the attribute representation. In principle, the attribute heads benefit from substantially more training signals than the rare classes alone provide, and the final class head can exploit attribute-level features as additional inductive bias.

We formalize this as Characteristic-Aware Hierarchical Detection (CHARD), which we evaluate under two integration strategies: (i) an end-to-end variant in which the hierarchical head replaces the standard classification head of RT-DETR (Section IV-B2), and (ii) an auxiliary-task variant in which an attribute head runs in parallel to a standard YOLO detection head and supervises the shared backbone through an additional loss (Section IV-B3). The auxiliary variant is backbone-agnostic, and we deliberately evaluate it on two backbones of differing capacity (YOLOv8-l and the stronger YOLOv11-l) to test whether the utility of characteristic supervision depends on backbone expressiveness.

1) HIERARCHICAL HEAD ARCHITECTURE

Let $\phi(x) \in \mathbb{R}^D$ denote the per-query (or per-anchor) feature vector produced by the backbone–neck network. The hierarchical head computes three parallel attribute prediction heads, each implemented as a single linear projection:

$$\hat{a}_w = \mathrm{softmax}(W_w \phi(x)), \quad W_w \in \mathbb{R}^{|\mathcal{W}| \times D}, \tag{1}$$

$$\hat{a}_s = \mathrm{softmax}(W_s \phi(x)), \quad W_s \in \mathbb{R}^{|\mathcal{S}| \times D}, \tag{2}$$

$$\hat{a}_p = \mathrm{softmax}\big(W_p \phi(x)\big), \quad W_p \in \mathbb{R}^{|\mathcal{P}| \times D}, \tag{3}$$

where $\mathcal{W} = \{2, 3, 4, \text{other}\}$, $\mathcal{S} = \{\text{small}, \text{medium}, \text{big}\}$, and $\mathcal{P} = \{\text{paddle}, \text{fossil}, \text{electric}, \text{human}, \text{animal}\}$.

To form a differentiable attribute representation, we compute the expected attribute embedding as a soft mixture over learned attribute prototypes:

$$\mathbb{E}[a|x] = \left[\, \hat{a}_w^\top E_w \parallel \hat{a}_s^\top E_s \parallel \hat{a}_p^\top E_p \,\right] \in \mathbb{R}^{3d_a}, \tag{4}$$

where $E_w \in \mathbb{R}^{|\mathcal{W}| \times d_a}$, $E_s \in \mathbb{R}^{|\mathcal{S}| \times d_a}$, $E_p \in \mathbb{R}^{|\mathcal{P}| \times d_a}$ are learned embedding matrices, $\parallel$ denotes concatenation, and $d_a = 64$ in our implementation.

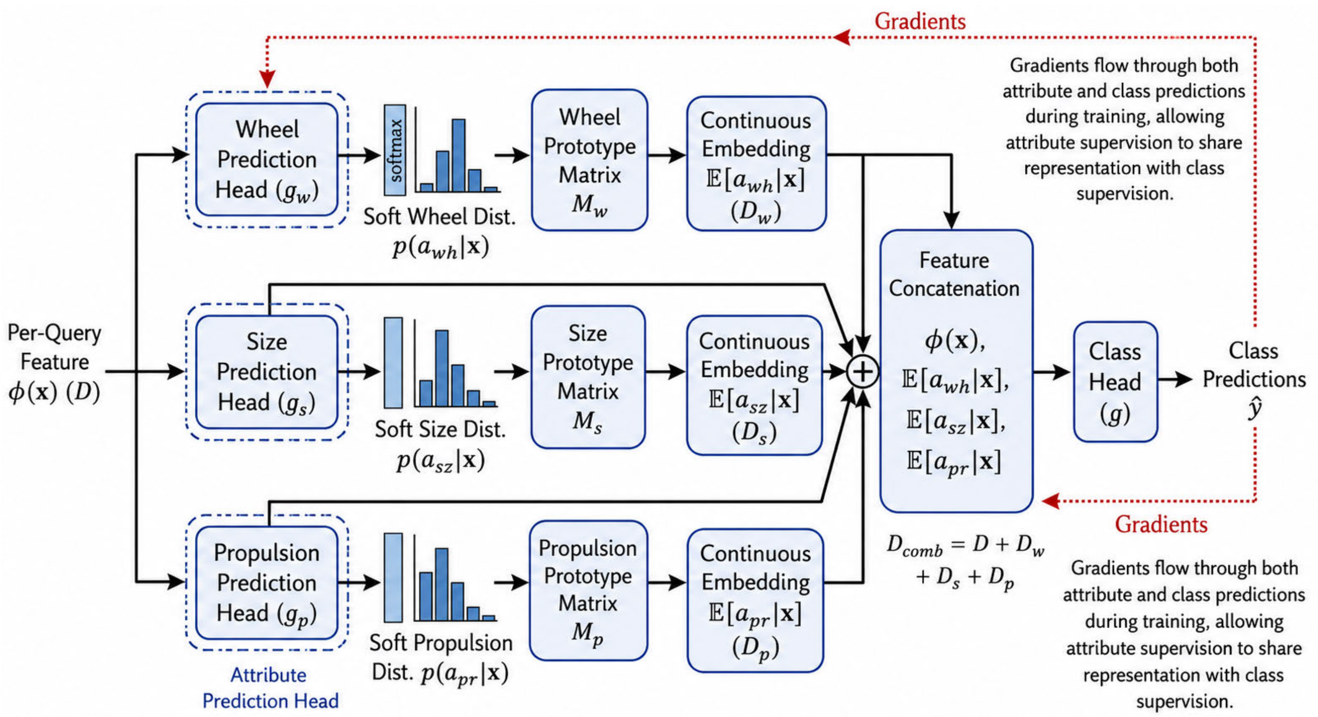

FIGURE 3. The CHARD hierarchical head. The per-query feature $\phi(x)$ is fed in parallel to three attribute prediction heads (wheel, size, propulsion). The resulting soft attribute distributions are converted to a continuous embedding $\mathbb{E}[a|x]$ via learned prototype matrices, which is concatenated with $\phi(x)$ and passed to a class head g. Gradients flow through both attribute and class predictions during training, allowing attribute supervision to share representation with class supervision.

The class head then operates on the concatenation of the original feature and the soft attribute embedding:

$$P(c \mid x) = \text{softmax}\Big(g\big(\phi(x),\ \mathbb{E}[a|x]\big)\Big), \tag{5}$$

where $g(\cdot,\cdot)$ is a two-layer MLP. The full architecture is depicted in Figure 3.

2) CHARD-RT-DETR VARIANT

In the end-to-end variant, the hierarchical head replaces the standard classification branch of RT-DETR. Object queries from the transformer decoder are passed through the hierarchical head defined in Equation 5, while box regression remains unchanged. The full pipeline is illustrated in Figure 4.

a: HUNGARIAN MATCHING WITH SIGMOID FOCAL COST

For each image, we compute the assignment between the Q object queries and N ground-truth objects by minimizing

$$\mathcal{C}_{ij} = \lambda^{m}_{\text{cls}}\mathcal{C}_{\text{cls}}(i,j) + \lambda_{\text{bbox}}\mathcal{C}_{\text{bbox}}(i,j) + \lambda_{\text{giou}}\mathcal{C}_{\text{giou}}(i,j), \tag{6}$$

where the classification cost uses the sigmoid focal formulation

$$\mathcal{C}_{\text{cls}}(i,j) = \alpha(1-p_{i,j})^{\gamma}\log p_{i,j} - (1-\alpha)p_{i,j}^{\gamma}\log(1-p_{i,j}), \tag{7}$$

with $p_{i,j}$ the sigmoid score of query i for ground-truth class j. The assignment is solved via the Hungarian algorithm [44].

b: COMPOSITE LOSS

The full training objective combines six terms:

$$\begin{aligned}\mathcal{L}_{\text{CHARD}} = \ &\lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{att}}\mathcal{L}_{\text{att}} + \lambda_{\text{con}}\mathcal{L}_{\text{con}} \\ &+ \lambda_{\text{dom}}\mathcal{L}_{\text{dom}} + \lambda_{\text{box}}\mathcal{L}_{\text{box}} + \lambda_{\text{giou}}\mathcal{L}_{\text{giou}}.\end{aligned} \tag{8}$$

The components are defined as follows. The classification loss $\mathcal{L}_{\text{cls}}$ is a sigmoid focal loss [24] applied to all Q queries, with unmatched queries assigned the all-zero target vector. The attribute loss $\mathcal{L}_{\text{att}}$ is the unweighted average of cross-entropy losses over the three attribute heads, computed only on matched queries:

$$\mathcal{L}_{\text{att}} = \frac{1}{3}\sum_{k\in\{w,s,p\}} \text{CE}\Big(\hat{a}_k,\ a_k^{\text{gt}}\Big). \tag{9}$$

The consistency loss $\mathcal{L}_{\text{con}}$ encourages agreement between the direct class prediction and the class distribution implied by attribute predictions through a Kullback–Leibler divergence

$$\mathcal{L}_{\text{con}} = \text{KL}\Bigg(P(c \mid x)\ \Big\|\ \sum_{a} P(c \mid a)\,P(a \mid x)\Bigg), \tag{10}$$

where $P(c \mid a)$ is a fixed lookup table derived from the taxonomy. The domain alignment loss $\mathcal{L}_{\text{dom}}$ regularizes the pooled image-level features toward a global exponential moving

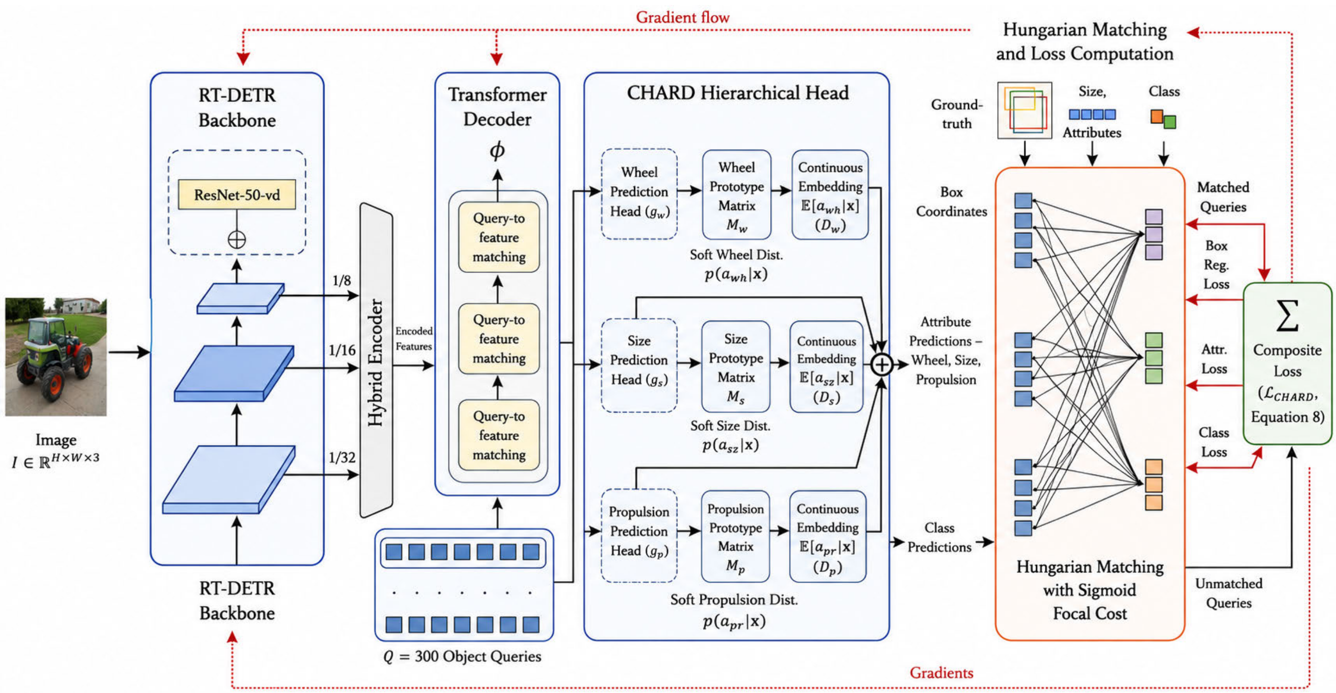

FIGURE 4. **CHARD-RT-DETR training pipeline. The RT-DETR backbone and hybrid encoder produce multi-scale features that feed the transformer decoder. Each of $Q = 300$ object queries is processed by the hierarchical head, yielding box coordinates, attribute predictions, and class predictions. Hungarian matching with sigmoid focal cost assigns queries to ground-truth targets, and the composite loss (Equation 8) is computed over matched pairs and unmatched queries.**

average of feature statistics, encouraging district-invariance:

$$\mathcal{L}_{\text{dom}} = \frac{1}{|D_B|} \sum_{d \in D_B} \|\mu_d - \mu_{\text{EMA}}\|_2^2 + \left\|\sigma_d^2 - \sigma_{\text{EMA}}^2\right\|_F^2, \tag{11}$$

where D_B is the set of districts represented in the current batch and μ_d, σ_d^2 are the within-district mean and variance. Finally, $\mathcal{L}_{\text{box}}$ and $\mathcal{L}_{\text{giou}}$ are the standard L_1 and GIoU losses [45] on matched query boxes. Loss weights are set to $\lambda_{\text{cls}} = 8$, $\lambda_{\text{att}} = 0.25$, $\lambda_{\text{con}} = 0.1$, $\lambda_{\text{dom}} = 0.05$, $\lambda_{\text{box}} = 5$, $\lambda_{\text{giou}} = 2$ following preliminary experiments. We additionally apply auxiliary versions of all losses to the outputs of intermediate decoder layers, averaged with weight $\lambda_{\text{aux}} = 1$.

3) CHARD-YOLO AUXILIARY VARIANT (YOLOv8-L AND YOLOv11-L)

The auxiliary variant, illustrated in Figure 5, leaves the base YOLO model and its loss largely unchanged, and instead introduces a parallel attribute prediction branch operating on the multi-scale FPN features. For each of the three FPN scales $\{P_3, P_4, P_5\}$ with channel counts C_3, C_4, C_5, the attribute branch applies a 3×3 convolution followed by a 1×1 projection to $|\mathcal{W}| + |\mathcal{S}| + |\mathcal{P}| = 4 + 3 + 5 = 12$ output channels. The result is reshaped to match the anchor layout used by the Detect head, producing per-anchor attribute logits. Because both YOLOv8-l and YOLOv11-l share this anchor-free FPN–Detect structure, the branch is architecturally identical on the two backbones; we therefore instantiate the variant on both (CHARD-YOLOv8 and CHARD-YOLOv11), holding the attribute branch, targets, and loss weight fixed so that any difference in outcome is attributable to backbone capacity alone.

We reuse YOLOv8's Task-Aligned Assigner [43] to identify foreground anchors $\mathcal{F}$. For each foreground anchor with assigned class c, attribute targets are derived from the class-to-attribute mapping (w_c, s_c, p_c). It is worth making the provenance of these targets explicit: BadODD contains no per-object attribute annotations, and CHARD requires none. The triple (w_c, s_c, p_c) is obtained by a fixed table lookup on the class label that the assigner has already assigned to the anchor, using the class-level taxonomy published with the dataset (Section III-F). Attribute supervision therefore adds no annotation cost, and transfers to any dataset whose classes can be placed in the same taxonomy. At inference the attribute branch is disabled entirely, so no attribute is ever predicted or evaluated. The attribute loss is the average over the three attribute heads of cross-entropy on foreground anchors only:

$$\mathcal{L}_{\text{att}}^{\text{aux}} = \frac{1}{3|\mathcal{F}|} \sum_{i \in \mathcal{F}} \sum_{k \in \{w,s,p\}} \text{CE}\left(\hat{a}_k^{(i)}, a_k^{(c_i)}\right). \tag{12}$$

The total training objective is the standard YOLO detection loss augmented with the attribute term:

$$\mathcal{L}_{\text{CHARD-YOLO}} = \mathcal{L}_{\text{YOLO}} + \lambda_{\text{attr}} \mathcal{L}_{\text{att}}^{\text{aux}}, \tag{13}$$

with $\lambda_{\text{attr}} = 0.5$ (and $\lambda_{\text{attr}} = 0$ for the no-attribute control variants). At inference time, the attribute branch is disabled, and the model behaves identically to the vanilla backbone, so this variant adds no runtime cost. This design isolates the contribution of attribute supervision: if the auxiliary task helps, it should manifest as improved class prediction on the same architecture and training pipeline that produces the corresponding YOLO baseline. Evaluating the identical design on YOLOv8-l and YOLOv11-l lets us test directly whether that contribution depends on the backbone.

V. EXPERIMENTAL SETUP

All experiments use the BadODD dataset described in Section III, in its standard configuration of 10,032 images partitioned into training (5,967), validation (2,032), and testing (2,033) splits in a 60:20:20 ratio. We did not modify the splits, ensuring direct comparability with the dataset paper [6] and any follow-up work using the standard partitioning.

We evaluate models under three complementary protocols:

- **Protocol 1 (P1) – standard random splits:** validation and test sets as released by the dataset authors. This produces the aggregate $\text{mAP}_{50:95}$ and mAP_{50} comparable to numbers in the original BadODD paper.
- **Protocol 2 (P2) – per-district stratified evaluation:** each validation image is grouped by its source district (derived from the filename district prefix), and detection metrics are computed independently per district. This protocol reveals geographic generalization patterns that aggregate metrics conceal.
- **Protocol 3 (P3) – per-frequency-group evaluation:** the 13 classes are partitioned into Head, Medium, and Tail groups based on their training-set frequency (Section III, Table 2), and group-level AP is reported as the unweighted mean of per-class AP within each group.

A. HARDWARE AND SOFTWARE

All experiments were conducted on a single NVIDIA RTX 3090 GPU with 24 GB of memory, running CUDA 13.0 with PyTorch 2.12 and Python 3.13. The four baseline detectors (YOLOv8-l, YOLOv10-l, YOLOv11-l, RT-DETR-l) were trained using the Ultralytics framework (version 8.4.52). The CHARD-RT-DETR variant was implemented using the HuggingFace Transformers RT-DETR base architecture (`PekingU/rtdetr_r50vd_coco_o365`) with a custom training loop in PyTorch. The CHARD-YOLO auxiliary variants (on the YOLOv8-l and YOLOv11-l backbones) were implemented by subclassing the Ultralytics DetectionModel and v8DetectionLoss to add the attribute head and auxiliary attribute loss. All training code, configuration files, and pre-trained checkpoints are released under an open-source license at the URL in our supplementary material.

B. TRAINING CONFIGURATION

This section details the training setup used throughout the experiments, including hardware specifications, hyperparameter settings, optimization strategies, data preprocessing procedures, and evaluation protocols. Providing these configurations ensures the reproducibility of the experiments and enables a fair comparison between the baseline models and the proposed CHARD framework.

1) BASELINE DETECTORS (ULTRALYTICS)

The four baseline detectors use identical optimization settings to ensure fair comparison: 100 training epochs with early stopping on patience 20, input resolution 640 × 640, batch size 32 for YOLOv8/v11-l, 24 for YOLOv10-l (whose dual-assignment head exceeds memory at 32), and 12 for RT-DETR-l (due to its higher memory footprint), with Ultralytics accumulating gradients to a nominal batch of 64 so that the effective optimization batch is identical across models; automatic mixed-precision (FP16) training, cosine learning rate schedule with 3-epoch linear warmup, initial learning rate 10^{-3} for YOLO models and 10^{-4} for RT-DETR-l, momentum 0.937, and weight decay 5×10^{-4}. All YOLO models use the Ultralytics default augmentation suite (mosaic, MixUp, HSV jitter, RandAugment, random horizontal flip with $p = 0.5$, scale and translation perturbations), with mosaic disabled in the final 10 epochs (`close_mosaic=10`) following standard YOLO recipes. Random seed is fixed to 0 for reproducibility. Training data is cached in RAM to minimize I/O overhead.

2) CHARD-RT-DETR

The custom variant is trained for 100 epochs at batch size 24 using AdamW with a base learning rate of 10^{-4} for the detector backbone and 5×10^{-4} for the hierarchical head (a 5× multiplier reflecting fresh initialization), weight decay 0.05, and a cosine schedule with 500-iteration linear warmup. Because CHARD-RT-DETR runs on a different implementation and backbone than the RT-DETR-l baseline (the HuggingFace ResNet-50-vd checkpoint with a custom BFloat16 training loop, versus the Ultralytics HGNetv2 baseline), the two batch sizes are not directly comparable; we set each to the largest that fits in the 24 GB of the RTX 3090 under its own memory profile. Gradients are clipped to L2-norm 0.5 during the first epoch and 1.0 thereafter. Training uses BFloat16 mixed precision (selected after FP16 caused intermittent overflow in the domain alignment EMA buffers). The hierarchical head's classification layer is initialized with a focal-loss prior bias ($\sigma^{-1}(0.01) \approx -4.595$) to prevent early training instability. Joint district-class repeat-factor sampling is applied to the training loader with thresholds $t_c = t_d = 10^{-3}$. The composite loss uses weights $\lambda_{\text{cls}} = 8$, $\lambda_{\text{att}} = 0.25$, $\lambda_{\text{con}} = 0.1$, $\lambda_{\text{dom}} = 0.05$, $\lambda_{\text{box}} = 5$, $\lambda_{\text{giou}} = 2$, and an auxiliary-decoder-layer weight $\lambda_{\text{aux}} = 1$. Augmentation is limited to horizontal flip, color jitter, and copy-paste of tail-class instances at probability

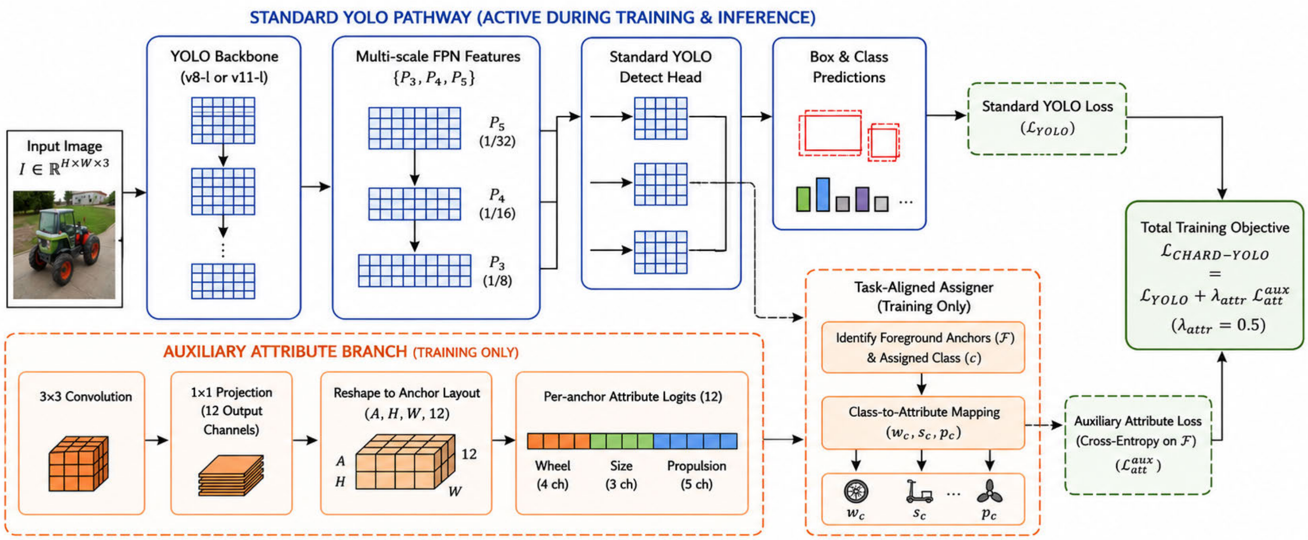

FIGURE 5. The CHARD-YOLO auxiliary variant. The standard YOLO pathway (backbone, FPN, Detect head, and box/class predictions, in blue) is unchanged and runs at both training and inference. During training only, an auxiliary attribute branch (orange) processes the same multi-scale FPN features through a 3 × 3 convolution and a 1 × 1 projection to twelve per-anchor logits, split across the three characteristic attributes (wheel, size, propulsion). YOLO's Task-Aligned Assigner selects foreground anchors and their assigned classes, from which the deterministic class-to-attribute mapping (w_c, s_c, p_c) provides cross-entropy targets. The total objective adds the auxiliary attribute loss to the standard YOLO loss, $\mathcal{L}_{\text{CHARD-YOLO}} = \mathcal{L}_{\text{YOLO}} + \lambda_{\text{attr}}\mathcal{L}_{\text{att}}^{\text{aux}}$ with $\lambda_{\text{attr}} = 0.5$ ($\lambda_{\text{attr}} = 0$ for the no-attribute controls). The attribute branch is discarded at inference, so the deployed model is identical to the vanilla backbone. The same design is applied to the YOLOv8-l and YOLOv11-l backbones.

0.2; mosaic and MixUp were attempted but caused training instability and were disabled.

3) CHARD-YOLO AUXILIARY VARIANTS (YOLOv8-L, YOLOv11-L)

Each auxiliary variant inherits its base detector's training configuration (YOLOv8-l or YOLOv11-l) described above without modification, with the single addition of the attribute branch and its associated loss term $\lambda_{\text{attr}}\mathcal{L}_{\text{att}}^{\text{aux}}$ with $\lambda_{\text{attr}} = 0.5$. No other hyperparameters are altered. For each backbone we also train a no-attribute control ($\lambda_{\text{attr}} = 0$) that carries the same added head but disables its loss, so that any observed performance change is attributable to the auxiliary task rather than the extra parameters. This isolates the contribution of attribute supervision on each backbone independently.

C. EVALUATION

For Protocols 1 and 2, all models are evaluated using the Ultralytics validation pipeline, which implements COCO-style mAP$_{50:95}$ at IoU thresholds $\{0.50, 0.55, \ldots, 0.95\}$ with maximum 300 detections per image. The CHARD-RT-DETR variant requires a separate evaluation pipeline because it uses sigmoid rather than the YOLO objectness pipeline; for it we use `torchmetrics` with the `pycocotools` backend and identical IoU thresholds, post-processing predictions with sigmoid scoring over 13 real classes (excluding the background dimension), top-300 selection per image, and score threshold 0.001. For Protocol 3, per-class AP from each Protocol-1 evaluation is averaged within frequency groups defined in Table 2. We report results on both validation and test splits throughout.

For Protocol 2, we stratify the released validation images by district based on filename prefixes (e.g., `dhaka_*`, `mymensingh_*`). Each district subset is treated as a standalone evaluation dataset: ground-truth labels remain the published BadODD annotations, and predictions are restricted to images in that district. Districts with fewer than 5 images are excluded from per-district reporting; all 9 districts in the validation set satisfy this threshold (Section VI-B).

D. MULTI-SEED AND CROSS-DATASET PROTOCOL

The single-run tables of Section VI fix all random seeds to 0 for exact reproducibility. To quantify training stochasticity and verify that our conclusions are not seed- or dataset-specific, we additionally conduct a multi-seed, multi-dataset study (Section VI-F): every architecture is retrained from scratch under three seeds $\{0, 1, 2\}$ on each of three datasets. Beyond BadODD we include two further Bangladeshi road-vehicle datasets, Poribohon-BD [40] (5,205 images) and Sorokh-Poth [41] (4,450 images), whose native class vocabularies we remap onto the 13-class BadODD taxonomy with a released conversion script, holding all optimization hyperparameters identical to the BadODD protocol. For this study we additionally evaluate CHARD on the stronger YOLOv11-l backbone and include no-attribute control variants (the CHARD architecture trained with $\lambda_{\text{attr}} = 0$) that isolate the effect of the attribute loss from that of the added head. Statistical comparisons pair runs by seed and use the paired t-test (primary) with the Wilcoxon signed-rank test (nonparametric confirmation), reported both within each dataset and over all nine matched runs combined.

VI. RESULTS

We report results under the three evaluation protocols defined in Section V: aggregate Protocol 1 on the standard splits, per-district Protocol 2 for geographic generalization, and per-frequency-group Protocol 3 for long-tail behavior. We then present qualitative comparisons on representative challenging cases.

A. AGGREGATE RESULTS (PROTOCOL 1)

Table 5 summarizes detection performance on the standard validation and test splits at seed 0. Among baselines, YOLOv11-l achieves the highest validation $mAP_{50:95}$ (0.4907), and the two YOLO leaders are effectively tied on test (YOLOv11-l 0.4254, YOLOv8-l 0.4208), with RT-DETR-l (0.4119) and YOLOv10-l (0.3961) trailing by 1–3 mAP points. The single-seed picture is already visibly noisy, however: on the test split the highest $mAP_{50:95}$ at this seed belongs not to any baseline but to the CHARD-YOLOv8 no-attribute *control* (0.4491), which the multi-seed analysis below shows to be a mid-pack model (0.4331 seed-averaged) rather than the leader a single run suggests; precisely the seed sensitivity that motivates Section VI-F. The custom CHARD-RT-DETR variant lags substantially (val mAP_{50} 0.5452 vs. RT-DETR-l 0.7469), reflecting the difficulty of optimizing a fresh hierarchical head jointly with a COCO-pretrained backbone using a custom training loop. Among the CHARD-YOLO variants, CHARD-YOLOv11 (attr) already outperforms all four plain baselines on every aggregate metric even at this single seed (val $mAP_{50:95}$ 0.4976, val mAP_{50} 0.7592, test mAP_{50} 0.6932); because these single-run values are the very quantities the remainder of this section shows to be seed-sensitive, we defer its definitive, seed-averaged evaluation to Section VI-F.

TABLE 5. Aggregate single-run detection performance on BadODD (Protocol 1) at the fixed seed 0. Every row is the seed-0 run of the multi-seed campaign (Section VI-F), so each value here is exactly one of the three points averaged in Table 8; CHARD-RT-DETR is the single custom-pipeline run. These single-run numbers are seed-sensitive; their seed-averaged, statistically tested counterparts appear in Section VI-F; and we defer definitive conclusions to that analysis. Bold marks the best value per column.

Model	Validation		Test	
	$mAP_{50:95}$	mAP_{50}	$mAP_{50:95}$	mAP_{50}
YOLOv8-l	0.4771	0.7233	0.4208	0.6828
YOLOv10-l	0.4712	0.7051	0.3961	0.6304
YOLOv11-l	0.4907	0.7356	0.4254	0.6555
RT-DETR-l	0.4732	0.7469	0.4119	0.6674
CHARD (RT-DETR)	0.2717	0.5452	0.2524	0.5113
CHARD-YOLOv8 (no attr)	0.4748	0.7055	**0.4491**	**0.6976**
CHARD-YOLOv8 (attr)	0.4834	0.7237	0.4250	0.6732
CHARD-YOLOv11 (no attr)	0.4866	0.7352	0.4101	0.6446
CHARD-YOLOv11 (attr)	**0.4976**	**0.7592**	0.4356	0.6932

B. GEOGRAPHIC DOMAIN GENERALIZATION (PROTOCOL 2)

Under Protocol 2, validation images are stratified by source district and detection metrics are computed independently per district. Detailed per-district results for all eight models, averaged over the three training seeds, are provided in Table S1 of the Supplementary Material. The results reveal substantial operational variance across regions. Every model performs best in Mymensingh (range 0.614–0.652) and worst in the cluster of Maowa (0.305–0.323), Sirajganj (0.294–0.325), and Sherpur (0.315–0.373). The gap between the easiest and hardest measurements reaches a factor of 2.22× (Mymensingh 0.652 vs. Sirajganj 0.294), and the within-model gap is remarkably stable across architectures, spanning only 2.02× to 2.12× for all eight models.

The most striking pattern is how little the choice of architecture matters relative to geography. The five easiest districts appear in *identical* rank order (Mymensingh, Rajshahi, Dhaka, Sylhet, Khulna) for all eight models, and within any district the spread between the best and worst architecture is typically 0.03–0.05 mAP (median 0.04), reaching 0.084 on the smallest district, Rajshahi, where RT-DETR-l's larger variance inflates the range; these spreads are comparable to or smaller than the seed-to-seed standard deviation in the harder districts (which reaches 0.063 on Sherpur). We therefore refrain from declaring per-district architectural winners: the apparent leader changes from district to district, and the margins are not separable from training noise at three seeds. What survives this scrutiny is the geographic effect itself, which is an order of magnitude larger and is reproduced by every architecture we evaluated. Figure 6 visualizes the resulting per-model distributions across districts.

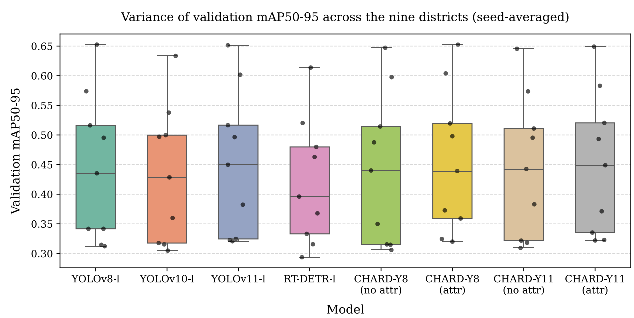

FIGURE 6. Distribution of seed-averaged validation $mAP_{50:95}$ across the nine geographic districts, one box per model (each point is one district). The distributions are nearly indistinguishable across the eight architectures and uniformly wide, showing that geographic variation dominates architectural variation (Section VII-C).

C. LONG-TAILED STRATIFICATION (PROTOCOL 3)

Protocol 3 isolates performance across frequency-based groupings: Head ($\geq$ 10% of training instances), Medium ($\geq 3\%$), and Tail ($< 3\%$). Table 6 reports the seed-averaged results for all eight primary models, plus the single-run CHARD-RT-DETR. Three patterns emerge.

First, CHARD-YOLOv11 (attr) attains the best or tied-best mean AP in every frequency group on both splits (it ties the YOLOv11-l baseline on the test Medium group at 0.524), and its advantage is largest exactly where it should be if characteristic supervision aids rare classes: on the Tail group it reaches 0.433 (val) and 0.319 (test), versus 0.420 and 0.280 for the YOLOv11-l baseline and 0.420 and 0.269 for its own no-attribute control. This is the group-level counterpart of the per-class decomposition in Figure 7. Second, every model exhibits a large validation-to-test drop on the Tail group (e.g., YOLOv8-l: 0.422 → 0.287) while Head and Medium remain stable across splits, and the Tail column also carries the largest seed-to-seed standard deviation (up to 0.028). Both reflect the extreme scarcity of tail instances in the test data, where construction_vehicle, train, and wheelchair together total only 55 instances (11, 16, and 28 respectively; Table 1); per-split Tail numbers are therefore inherently high-variance, a measurement artifact rather than a property of any model. Third, the custom CHARD-RT-DETR run trails every YOLO-family model in all groups (Tail 0.177 val, 0.122 test), consistent with the joint-optimization difficulty discussed in Section VII-B rather than with any deficiency of the characteristic taxonomy itself. Notably, CHARD-YOLOv8 (attr) is, under three seeds, statistically indistinguishable from vanilla YOLOv8-l on every group, confirming that the auxiliary attribute head neither helps nor harms this backbone.

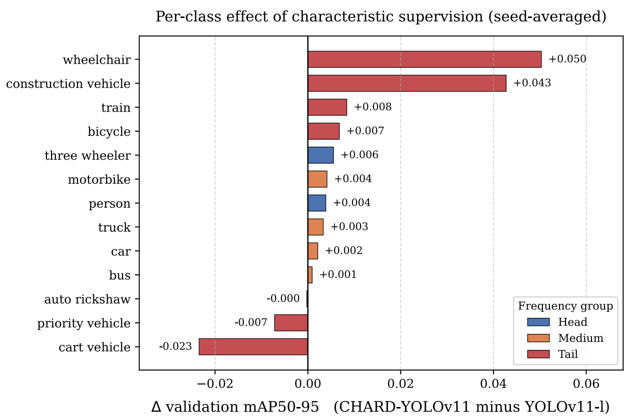

FIGURE 7. Per-class effect of characteristic supervision on the YOLOv11-l backbone: change in validation $mAP_{50:95}$ of CHARD-YOLOv11 (attr) relative to the YOLOv11-l baseline, averaged over three seeds and colored by frequency group. Head and Medium classes are essentially unchanged (≤ 0.006), while the largest movements occur on the rarest classes in both directions: wheelchair (+0.050) and construction_vehicle (+0.043) improve markedly, whereas cart_vehicle regresses (−0.023). The net effect on the Tail group is positive, but the per-class spread illustrates the high variance of tail estimates discussed in Section VII-D. This figure and Table 7 are computed from the same seed-averaged per-class records; the deltas shown here use full-precision AP and may therefore differ by up to 0.001 from the difference of the three-decimal entries in Table 7.

D. PER-CLASS BREAKDOWN

Table 7 reports seed-averaged per-class validation $mAP_{50:95}$ for all eight primary models and the single-run CHARD-RT-DETR. Head classes (auto_rickshaw, person, three_wheeler) and structurally distinctive Medium classes (bus, car, truck) achieve mAP above 0.5 for most architectures. The table also confirms two effects already noted at the group level. The CHARD-YOLOv11 (attr) variant is the top model on the largest number of classes (bus, person, three_wheeler, truck, among others), consistent with its group-level lead. Tail classes show wide spread and their per-class winners are erratic: construction_vehicle and wheelchair are topped by RT-DETR-l (0.346 and 0.566), and `train` by YOLOv8-l (0.843). These three maxima should not be over-interpreted, for two reasons. First, the classes carry only 6, 5, and 27 validation instances respectively (Table 1), so a single detection moves the score by a large fraction; the seed-to-seed standard deviation on them reaches 0.078, 0.019, and 0.089 for CHARD-YOLOv11 (attr), which is larger than most of the between-model gaps, and individual seeds of CHARD-YOLOv11 (attr) reach 0.352 on construction_vehicle, above the RT-DETR-l mean. Second, and more importantly, the controlled comparison runs the other way: on all three of these classes CHARD-YOLOv11 (attr) exceeds both its own YOLOv11-l baseline (0.269 vs. 0.226, 0.838 vs. 0.829, and 0.508 vs. 0.458) and its matched no-attribute control, which is the contrast that isolates the effect of characteristic supervision. The apparent RT-DETR-l advantage on two rare classes is also not accompanied by competitive overall accuracy, and the Tail group as a whole, which is the statistically meaningful unit here, is led by CHARD-YOLOv11 (attr) on both splits (Table 6). The custom CHARD-RT-DETR run is uniformly the weakest row, in line with its aggregate result. Notably, averaged over seeds CHARD-YOLOv8 (attr) equals or exceeds vanilla YOLOv8-l on ten of thirteen classes (all but bus, train, and wheelchair), confirming that the auxiliary head does not degrade the backbone and that any per-class deficits seen at a single seed are seed artifacts.

E. QUALITATIVE VISUAL ANALYSIS

We compare our reference baseline (YOLOv11-l, co-leading with YOLOv8-l) against the best overall model (CHARD-YOLOv11, attr) qualitatively, using two complementary figures placed where the corresponding claims are made. Throughout, predicted detections are drawn as solid green boxes (with confidence), and ground-truth objects a model fails to detect are marked with red dashed circles, so the differences between the two models are directly legible. Figure 8 first establishes that both models are robust across the full range of BadODD imaging conditions: dense daytime urban traffic (Dhaka), low-light night driving (Chittagong), and a high-speed expressway scene with small distant objects (Maowa). Under normal lighting, both models reliably localize the head classes (person, auto_rickshaw, three_wheeler); at night, both degrade gracefully, detecting the well-lit foreground vehicles while missing a few heavily shadowed pedestrians; and on the clean expressway scene, the two are near-identical, both localizing the distant bus and roadside truck. Even here, however, the benefit of characteristic supervision is already visible on rare classes: in the Dhaka scene, CHARD-YOLOv11 recovers a `cart_vehicle` (conf. 0.38) that the baseline misses.

TABLE 6. Per-frequency-group $mAP_{50:95}$ on BadODD validation and test. The eight primary models are averaged over three seeds (mean ± std); CHARD-RT-DETR is a single custom-pipeline run and is reported without a standard deviation. Head (AP_H)/Medium (AP_M)/Tail (AP_T) groups follow Table 2. Bold marks the best mean per column.

Model	Validation			Test		
	AP_H	AP_M	AP_T	AP_H	AP_M	AP_T
YOLOv8-l	0.568 ± 0.001	0.513 ± 0.002	0.422 ± 0.013	0.578 ± 0.001	0.520 ± 0.002	0.287 ± 0.012
YOLOv10-l	0.562 ± 0.002	0.499 ± 0.005	0.411 ± 0.020	0.571 ± 0.001	0.508 ± 0.002	0.230 ± 0.018
YOLOv11-l	0.571 ± 0.001	0.514 ± 0.002	0.420 ± 0.017	0.580 ± 0.001	0.524 ± 0.004	0.280 ± 0.014
RT-DETR-l	0.522 ± 0.011	0.480 ± 0.016	0.425 ± 0.010	0.529 ± 0.014	0.492 ± 0.016	0.297 ± 0.020
CHARD-RT-DETR	0.389	0.326	0.177	0.401	0.338	0.122
CHARD-Y8 (no attr)	0.565 ± 0.000	0.510 ± 0.004	0.406 ± 0.005	0.575 ± 0.001	0.516 ± 0.004	0.307 ± 0.028
CHARD-Y8 (attr)	0.568 ± 0.002	0.516 ± 0.002	0.422 ± 0.012	0.577 ± 0.000	0.523 ± 0.004	0.279 ± 0.012
CHARD-Y11 (no attr)	0.572 ± 0.001	0.512 ± 0.002	0.420 ± 0.008	0.580 ± 0.001	0.522 ± 0.005	0.269 ± 0.019
CHARD-Y11 (attr)	$\mathbf{0.574 \pm 0.003}$	$\mathbf{0.517 \pm 0.002}$	$\mathbf{0.433 \pm 0.014}$	$\mathbf{0.581 \pm 0.005}$	$\mathbf{0.524 \pm 0.001}$	$\mathbf{0.319 \pm 0.013}$

TABLE 7. Per-class validation $mAP_{50:95}$ on BadODD, with models as rows and classes as columns. The eight primary models are seed-averaged over three seeds; CHARD-RT-DETR is a single custom-pipeline run. Bold marks the best value per class. Per-class values for the three rarest classes (Const, Train, W.chair) rest on very few validation instances and are individually high-variance (Section VII-D).

Model	A.rick	Bicyc	Bus	Car	Cart	Const	M.bike	Person	Prio	3-whl	Train	Truck	W.chair
YOLOv8-l	0.639	0.325	0.517	0.561	0.146	0.219	0.428	0.446	0.511	0.618	**0.843**	0.546	0.488
YOLOv10-l	0.636	0.317	0.487	0.554	0.164	0.217	0.428	0.442	0.499	0.609	0.839	0.524	0.433
YOLOv11-l	**0.644**	0.317	0.519	0.562	0.168	0.226	0.429	0.448	0.525	0.622	0.829	0.548	0.458
RT-DETR-l	0.596	0.278	0.463	0.536	0.176	**0.346**	0.408	0.404	0.517	0.566	0.669	0.513	**0.566**
CHARD-RT-DETR	0.456	0.162	0.338	0.363	0.059	0.015	0.239	0.300	0.355	0.412	0.398	0.363	0.074
CHARD-Y8^{na}	0.637	0.319	0.512	0.560	0.159	0.204	0.427	0.445	**0.533**	0.612	0.825	0.540	0.395
CHARD-Y8^{a}	0.641	**0.325**	0.516	**0.565**	**0.195**	0.232	**0.435**	0.446	0.517	0.618	0.840	0.548	0.424
CHARD-Y11^{na}	0.642	0.313	0.516	0.562	0.154	0.234	0.428	0.449	0.512	0.625	0.828	0.544	0.483
CHARD-Y11^{a}	0.644	0.323	**0.520**	0.564	0.145	0.269	0.433	**0.452**	0.518	**0.627**	0.838	**0.552**	0.508

The differences between the models sharpen on genuinely challenging cases, which we examine alongside the quantitative tail-class analysis in Section VII-B (Figure 9).

F. MULTI-SEED STABILITY, STATISTICAL SIGNIFICANCE, AND CROSS-DATASET GENERALIZATION

The aggregate comparisons in Section VI-A are single-run point estimates, and several of the inter-model gaps are of the same order as the training stochasticity inherent to detector optimization. To establish which of the reported differences are reproducible rather than seed artifacts, and to test whether our conclusions transfer beyond BadODD, we conducted an extended empirical study spanning three axes: (i) **three random seeds** $\{0, 1, 2\}$ per configuration, retraining every model from scratch; (ii) **two additional Bangladeshi road-vehicle datasets** beyond BadODD, namely Poribohon-BD [40] (5,205 images, 10 native classes) and Sorokh-Poth [41] (4,450 images), both remapped onto a common 13-class taxonomy compatible with BadODD; and (iii) **eight model variants**, adding CHARD-YOLOv11 (the hierarchical attribute head grafted onto the stronger YOLOv11-l backbone) and no-attribute control variants that instantiate the CHARD architecture with the attribute loss disabled ($\lambda_{\text{attr}} = 0$). This yields $8 \times 3 \times 3 = 72$ independently trained models. All runs use the identical optimization protocol of Section V; the two supplementary datasets are converted with a public script released alongside our code. This design directly addresses the concern that small performance gaps on a modestly sized dataset may be unstable, by quantifying the seed variance explicitly and by verifying that the ranking of architectures is preserved across three independent datasets.

1) STABILITY ACROSS SEEDS AND DATASETS

Table 8 reports the mean and standard deviation of test $mAP_{50:95}$ across the three seeds for every model on all three datasets. The most important observation is that *the seed variance is small*: the standard deviation is below 0.010 mAP for 20 of the 24 model×dataset cells, and never exceeds 0.014 (excluding RT-DETR-l on Sorokh-Poth, whose larger spread we discuss below). This tight concentration confirms that the detectors converge to consistent operating points and that the aggregate metrics in Table 5 are reliable rather than lucky draws. The relative ordering of architectures is also strikingly consistent across the three datasets: the YOLOv11-l and YOLOv8-l families occupy the top tier on all three, YOLOv10-l trails them, and RT-DETR-l is consistently weakest, indicating that our architectural conclusions are not idiosyncratic to BadODD.

2) STATISTICAL SIGNIFICANCE

Before analyzing CHARD, we first verify the robustness of the baseline ranking. Pooling the nine matched (dataset, seed) runs, YOLOv11-l significantly outperforms YOLOv10-l ($\Delta = +0.017$, paired t-test $p_t = 0.005$) and RT-DETR-l ($\Delta = +0.068$, $p_t = 0.002$), while remaining statistically indistinguishable from YOLOv8-l ($\Delta = +0.002, p_t = 0.41$).

FIGURE 8. Qualitative comparison across three representative *imaging conditions* (columns): dense daytime urban traffic, low-light night driving, and a high-speed expressway scene. Top row: YOLOv11-l (reference baseline); bottom row: CHARD-YOLOv11 (attr) (best overall model). Solid green boxes are predicted detections (with confidence); red dashed circles mark undetected ground-truth objects. Both models are robust across conditions and near-identical on the clean expressway scene.

TABLE 8. Multi-seed stability of test $mAP_{50:95}$ (mean ± standard deviation over three seeds {0, 1, 2}) across three Bangladeshi road-vehicle datasets. All runs, including the single-seed entries of Table 5, use the identical protocol of Section V (100 epochs maximum, early stopping at patience 20). Best result per dataset in **bold**. CHARD-YOLOv11 (attr) attains the best BadODD result of any model.

Model	BadODD	Poribohon-BD	Sorokh-Poth
YOLOv8-l	0.4258 ± 0.0054	0.6133 ± 0.0041	0.7708 ± 0.0048
YOLOv10-l	0.3944 ± 0.0092	0.6060 ± 0.0109	0.7647 ± 0.0048
YOLOv11-l	0.4245 ± 0.0059	0.6150 ± 0.0048	$\mathbf{0.7775 \pm 0.0088}$
RT-DETR-l	0.4105 ± 0.0120	0.5394 ± 0.0031	0.6631 ± 0.0265
CHARD-YOLOv8 (no attr)	0.4331 ± 0.0139	0.6080 ± 0.0008	0.7685 ± 0.0054
CHARD-YOLOv8 (attr)	0.4229 ± 0.0058	0.6119 ± 0.0033	0.7684 ± 0.0017
CHARD-YOLOv11 (no attr)	0.4187 ± 0.0090	$\mathbf{0.6150 \pm 0.0048}$	0.7771 ± 0.0096
CHARD-YOLOv11 (attr)	$\mathbf{0.4426 \pm 0.0069}$	0.6128 ± 0.0071	0.7747 ± 0.0087

Thus, the multi-seed evaluation confirms YOLOv11-l and YOLOv8-l as the two strongest baseline architectures on BadODD.

Table 10 evaluates whether CHARD changes detector accuracy beyond normal training variability. Each comparison matches a CHARD model with its corresponding baseline across the three training seeds on the BadODD test split. From the data, two conclusions emerge. First, disabling the attribute loss while retaining the additional prediction head produces no measurable effect on either YOLOv8-l or YOLOv11-l ($p_t = 0.56$ and 0.36), demonstrating that the added parameters alone do not influence detection accuracy. Second, the impact of characteristic supervision is architecture-dependent. On YOLOv8-l, the auxiliary task is statistically neutral ($\Delta = -0.003$, $p_t = 0.52$), whereas on YOLOv11-l it consistently improves performance ($\Delta = +0.018$). Although the comparison with the plain YOLOv11-l baseline is marginal under the conventional 5% threshold ($p_t = 0.052$), the matched comparison against the no-attribute control isolates the contribution of characteristic supervision and yields a larger, statistically significant improvement ($\Delta = +0.024$; the three per-seed paired differences are all positive at +0.026, +0.021, and +0.025; 95% CI [+0.018, +0.030]; paired t-test $p_t = 0.003$; Cohen's $d_z \approx 10$, reflecting the very tight seed agreement). Because Table 10 reports $m = 6$ paired hypotheses, we verified that this result survives a conservative Bonferroni correction over that family (unadjusted $p_t = 0.003$; adjusted $p = 0.018 < 0.05$). With only three seeds the normality of the paired differences cannot be assessed reliably, so we base the claim on the exact per-seed differences and the confidence interval rather than on the nonparametric Wilcoxon signed-rank test, which at $n = 3$ cannot fall below its minimum two-sided value of $p_W = 0.25$. This indicates that the observed gain is attributable to the characteristic-aware supervision itself rather than to the additional prediction head. Finally, the custom CHARD-RT-DETR model underperforms RT-DETR-l by 0.158 $mAP_{50:95}$; since only a single training run was available for this custom

TABLE 9. Multi-seed stability of *validation* $mAP_{50:95}$ (mean ± standard deviation over three seeds {0, 1, 2}, same protocol as Table 8) across the three datasets. Best result per dataset in bold. CHARD-YOLOv11 (attr) attains the best BadODD validation result (0.4916), the seed-averaged counterpart of its single-seed value in Table 5.

Model	BadODD	Poribohon-BD	Sorokh-Poth
YOLOv8-l	0.4835 ± 0.0057	0.6532 ± 0.0041	0.7772 ± 0.0019
YOLOv10-l	0.4731 ± 0.0114	0.6450 ± 0.0042	0.7760 ± 0.0023
YOLOv11-l	0.4841 ± 0.0073	0.6596 ± 0.0072	0.7843 ± 0.0011
RT-DETR-l	0.4644 ± 0.0120	0.5822 ± 0.0064	0.6628 ± 0.0333
CHARD-YOLOv8 (no attr)	0.4746 ± 0.0011	0.6526 ± 0.0011	0.7765 ± 0.0025
CHARD-YOLOv8 (attr)	0.4847 ± 0.0050	0.6568 ± 0.0024	0.7792 ± 0.0008
CHARD-YOLOv11 (no attr)	0.4837 ± 0.0041	**0.6596 ± 0.0072**	**0.7846 ± 0.0013**
CHARD-YOLOv11 (attr)	**0.4916 ± 0.0053**	0.6578 ± 0.0026	0.7835 ± 0.0016

implementation, no statistical test can be performed, and this result is reported for qualitative comparison only.

TABLE 10. Paired significance tests on the BadODD test $mAP_{50:95}$ over three training seeds ($n = 3$). The upper block compares each CHARD variant with its corresponding base detector; the lower block compares each attribute variant against its matched no-attribute control, the contrast that isolates the effect of characteristic supervision. Δ denotes the mean paired difference (first model minus second), and p_t is the paired t-test.

Comparison	Δ	p_t
CHARD-YOLOv8 (no attr) vs. YOLOv8-l	+0.0073	0.560
CHARD-YOLOv8 (attr) vs. YOLOv8-l	−0.0030	0.518
CHARD-YOLOv11 (no attr) vs. YOLOv11-l	−0.0057	0.358
CHARD-YOLOv11 (attr) vs. YOLOv11-l	+0.0182	0.052
CHARD-YOLOv8 (attr) vs. CHARD-YOLOv8 (no attr)	−0.0102	0.292
CHARD-YOLOv11 (attr) vs. CHARD-YOLOv11 (no attr)	+0.0239	**0.003**

3) WHEN CHARACTERISTIC SUPERVISION HELPS: A REFINED VIEW

The statistical analysis reveals that the effectiveness of CHARD is not universal but depends on the representational capacity of the underlying detector. On the YOLOv8-l backbone, characteristic-aware supervision provides no measurable benefit beyond the primary detection objective, suggesting that the auxiliary task is largely redundant for a capacity-limited model. In contrast, the stronger YOLOv11-l backbone consistently benefits from the same supervision, indicating that additional representational capacity is required to exploit the auxiliary characteristic information effectively.

A per-frequency-group analysis further clarifies where these improvements originate. CHARD-YOLOv11 (attr) increases Tail-group test AP to 0.319, compared with 0.269 for the matched no-attribute control and 0.280 for the vanilla YOLOv11-l baseline, while Head and Medium performance remain essentially unchanged. To confirm that this tail effect is not merely a by-product of the aggregate test, we ran a paired t-test on the Tail-group AP *itself*, matched by seed. Against the no-attribute control the three per-seed Tail-AP differences are all positive (+0.053, +0.043, +0.055; mean +0.050, 95% CI [+0.034, +0.067], $p_t = 0.006$), so the tail improvement is statistically significant relative to the control; against the plain YOLOv11-l baseline the mean gain is larger in absolute terms (+0.039) but only marginal ($p_t = 0.062$), mirroring the aggregate result. We therefore restrict the "statistically significant tail-class improvement" claim to the controlled (attribute-versus-no-attribute) comparison. Thus, the overall improvement is driven almost entirely by the rare categories rather than by gains on already well-represented classes. Given the extreme class imbalance of BadODD, where the most frequent class contains more than 800 times as many training instances as the rarest, this localized improvement contributes only modestly to the aggregate $mAP_{50:95}$ despite representing the primary objective of the CHARD framework.

These findings align closely with the motivation behind the characteristic taxonomy. By encouraging related vehicle categories to share supervision through common attributes, CHARD provides an additional learning signal precisely where class-level supervision is weakest. The absence of measurable changes on Head and Medium classes further suggests that the auxiliary supervision does not simply improve the detector uniformly, but preferentially strengthens the representation of under-represented categories. Consequently, CHARD-YOLOv11 (attr) achieves the highest overall performance among all evaluated models, reaching seed-averaged $mAP_{50:95}$ values of 0.4916 on the validation set (Table 9) and 0.4426 on the BadODD test set (Table 8).

Figure 9 provides qualitative evidence supporting this interpretation. Across three challenging scenes, CHARD-YOLOv11 consistently recovers rare, distant, or heavily occluded objects that are missed by the YOLOv11-l baseline. In the Mymensingh scene, it detects multiple overlapping `priority_vehicle` instances that remain undetected by the baseline. In the Sherpur scene, it successfully identifies partially occluded three-wheelers and an auto-rickshaw, while in the Rajshahi scene it more reliably localizes small, distant tail-class objects. These examples visually corroborate the quantitative Tail-group improvements while preserving identical inference complexity,

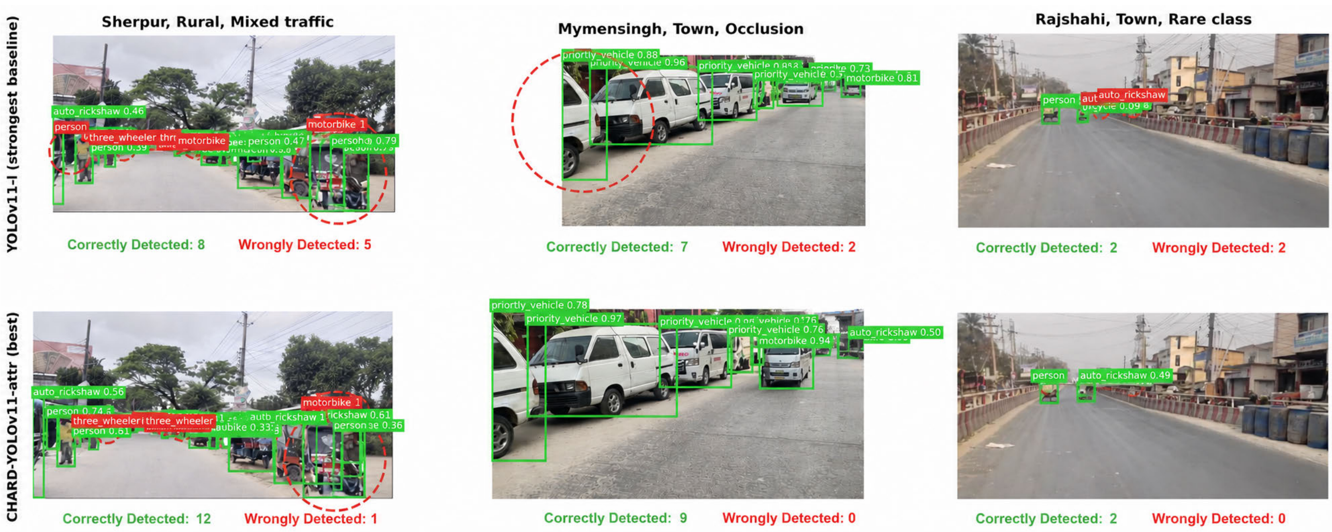

FIGURE 9. **Qualitative comparison on three *challenging* scenes (columns): a rural scene with dispersed small vehicles (Sherpur), a heavily occluded town scene (Mymensingh), and a scene with rare tail-class instances (Rajshahi). Top row: YOLOv11-l (reference baseline); bottom row: CHARD-YOLOv11 (attr) (best overall model). Solid green boxes are predicted detections (with confidence); red dashed circles mark undetected ground-truth objects. CHARD-YOLOv11 recovers occluded and rare tail-class objects that the baseline misses (most visibly the overlapping *priority_vehicle* instances in the Mymensingh scene), visually corroborating its tail-class improvement, which is statistically significant against its no-attribute control (Section VI-F).**

since the auxiliary characteristic branch is discarded during inference.

4) CROSS-DATASET GENERALIZATION OF RT-DETR

The three-dataset design also surfaces a finding invisible to any single-dataset study: RT-DETR-l generalizes markedly worse than the YOLO family as the data distribution shifts. Its deficit relative to YOLOv11-l widens from 0.014 mAP on BadODD to 0.076 on Poribohon-BD and 0.114 on Sorokh-Poth, and it loses on eight of the nine matched runs ($p_W = 0.008$; the sole exception is BadODD seed 2). RT-DETR-l also shows the largest seed variance (± 0.027 on Sorokh-Poth). For practitioners deploying on emerging-market road imagery, the anchor-free YOLO detectors are thus not only faster (Section VI-G) but also more robust to dataset shift.

G. INFERENCE EFFICIENCY AND REAL-TIME FEASIBILITY

Road-scene perception is latency-critical, so aggregate accuracy must be read alongside inference cost. We measured single-image latency under a controlled protocol: a fixed 640×640 input, batch size 1, 50 warm-up iterations followed by 300 timed iterations with CUDA event timing on the RTX 3090. These figures time the *model forward pass only*. At half precision every detector operates comfortably in real time, with mean (p95) forward-pass latency of 15.77 ms (16.94) for RT-DETR-l, 12.96 ms (13.29) for YOLOv11-l, 12.49 ms (12.70) for YOLOv10-l, and 7.75 ms (8.06) for YOLOv8-l, i.e. 63–129 FPS with tight tails. Adding end-to-end overhead (image decoding/resizing and, for the YOLO detectors, NMS post-processing) raises the per-image total by roughly 1 ms in our evaluation harness (e.g., YOLOv8-l 6.8 ms forward → 7.8 ms end-to-end), so the end-to-end rates remain well above the 30 FPS cadence of typical automotive and smartphone cameras. The accuracy comparisons throughout this paper are therefore drawn from a set of models that are all deployable on this class of GPU; the choice among them is an accuracy and robustness decision, not a feasibility one.

The inference-time implications for our proposed framework are especially favorable, and they follow directly from its design rather than from an empirical accident. Because the CHARD attribute branch is a training-only auxiliary head that is detached from the computational graph at inference (Section IV-B3), the deployed CHARD-YOLO model executes exactly the operations of its vanilla backbone. The compute profile confirms this: CHARD-YOLOv11 (attr) has *identical* inference complexity to YOLOv11-l at 87.3 GFLOPs, and the additional attribute-head parameters (which raise the stored checkpoint from 25.3M to 30.6M) are never evaluated at test time. The measured latencies bear this out, differing from the corresponding baselines by under 2%: 13.06 ms for CHARD-YOLOv11 (attr) versus 12.96 ms for YOLOv11-l (0.7%), and 7.88 ms for CHARD-YOLOv8 (attr) versus 7.75 ms for YOLOv8-l (1.7%). These 0.10–0.13 ms gaps fall within run-to-run timing jitter and do not reflect real added cost. The practical consequence is central to the value of the framework: the tail-class accuracy gain of CHARD-YOLOv11 (statistically significant against its no-attribute control; Section VI-F) is obtained at *zero* deployment overhead. Characteristic supervision therefore behaves as a pure training-time regularizer, purchasing accuracy on rare classes during optimization and then vanishing, so that the deployed detector is as fast as the baseline a practitioner would otherwise ship. This clean decoupling of a training-time accuracy benefit from inference cost is, in our view, the most deployment-relevant property of CHARD: unlike heavier architectural changes that trade latency for accuracy, it improves the long tail for free at run time. We measure all latencies on a single RTX 3090, which serves as a workstation- or datacenter-class upper bound on achievable speed; our real-time claim is therefore scoped to this class of GPU. Profiling these detectors on embedded automotive hardware (e.g., an NVIDIA Jetson Orin with TensorRT), together with a full end-to-end latency budget including decoding and NMS reported with p95 tails, is the necessary next step for a genuine on-vehicle deployment claim and is left to future work.

VII. DISCUSSION

The empirical results above support four principal observations that we now examine in turn: the strength of YOLOv11-l as a baseline, the backbone-dependent behavior of CHARD as a multi-task framework, the geographic variance in detection difficulty within Bangladesh, and the structural instability of tail-class metrics. Throughout, we draw on the multi-seed and cross-dataset evidence of Section VI-F to separate reproducible effects from training noise.

A. BEST BASELINE: YOLOv11-L

YOLOv11-l is one of the two co-leading baselines in our study (statistically tied with YOLOv8-l), achieving seed-averaged validation $mAP_{50:95}$ of 0.4841 and test $mAP_{50:95}$ of 0.4245 (Table 8); we adopt it as our reference baseline because it is the CHARD backbone. Its margin over YOLOv8-l is negligible at the aggregate level (within 0.001$mAP_{50:95}$ on both splits, with YOLOv8-l in fact nominally ahead on test) and, as the multi-seed analysis shows, not statistically separable from training noise; the two occupy the same top tier, and YOLOv11-l's durable advantages are over YOLOv10-l and RT-DETR-l, both of which it beats significantly (Section VI-F). This tier ordering is consistent at the frequency-group level (Table 6): YOLOv11-l and YOLOv8-l are effectively tied on the Tail group (test 0.280 vs. 0.287) while both exceed YOLOv10-l by roughly 0.05. We caution that per-split Tail comparisons between the two leaders are not meaningful given the group's high variance, and note that the overall best Tail performance belongs not to any baseline but to CHARD-YOLOv11 (attr). At the per-district level (Table S1, Supplementary Material) no architecture is a consistent winner once seeds are averaged, so we base this ranking on the significant aggregate separations rather than on geographic or tail-group wins. We attribute the gains over YOLOv10-l to the C3k2 block, which provides richer feature aggregation than the C2f modules of YOLOv8 at comparable computational cost, and to the retention of standard non-maximum suppression at inference. YOLOv10-l's removal of NMS through dual-label assignment reduces latency but appears to suffer in the dense, heavily occluded traffic that characterizes the BadODD dataset, where overlapping detections at high intersection-over-union are common.

B. WHEN CHARACTERISTIC SUPERVISION HELPS: A BACKBONE-DEPENDENT EFFECT

Our multi-seed analysis (Section VI-F) yields a more nuanced picture of the CHARD framework than any single run can provide, and reconciles an apparent negative result with a genuinely positive one. The central lesson is that the usefulness of characteristic-aware supervision is *backbone-dependent*, governed by a competition between two forces: the redundancy of the deterministic taxonomy, and the capacity of the backbone to convert the auxiliary gradient into better tail-class features. Throughout, we use "capacity" to mean representational expressiveness rather than raw model size: YOLOv11-l is in fact the *smaller* model (25.3M parameters, 86.6 GFLOPs versus YOLOv8-l's 43.7M and 165.4), and its advantage comes from the more effective C3k2 feature aggregation, not from parameter count.

Consider first the redundancy force, which dominates on the YOLOv8-l backbone. In BadODD the mapping from class to attribute triple is deterministic and many-to-one (e.g., `car` $\mapsto$ (4)-wheel, medium, fossil); `auto_rickshaw` $\mapsto$ (3)-wheel, medium, fossil)). It is in fact a strict coarsening of the label space: the 13 classes collapse onto only 10 distinct attribute triples, with `car`, `truck`, and `priority_vehicle` all sharing (4)-wheel, medium, fossil). Because the class label completely determines the attribute label, supervising both adds no orthogonal information and effectively duplicates the classification objective on a coarser label space; the auxiliary loss then competes with the primary classification and box-regression objectives for representational capacity, a form of gradient interference [37]. On YOLOv8-l this competition leaves performance statistically unchanged: across three seeds the attribute variant differs from vanilla YOLOv8-l by only -0.003 test $mAP_{50:95}$ ($p_t = 0.52$) and from its own no-attribute control by -0.010 ($p_t = 0.29$), both well within seed noise (Table 10). As a cautionary illustration of the same seed sensitivity, at seed 0 the CHARD-YOLOv8 no-attribute control posts the single highest test $mAP_{50:95}$ of any model (0.4491, Table 5), yet its seed-averaged value is an unremarkable 0.4331; sub-2-point gaps on a dataset of this size must therefore be interpreted through multi-seed statistics rather than single runs.

The opposing force, productive use of the auxiliary gradient, emerges on the stronger YOLOv11-l backbone. There, the same characteristic supervision becomes beneficial: CHARD-YOLOv11 (attr) achieves the highest BadODD test $mAP_{50:95}$ of any model (0.4426), significantly exceeding its no-attribute control ($\Delta = +0.024$, $p_t = 0.003$; Bonferroni-adjusted $p = 0.018$ over the six comparisons of Table 10; Section VI-F). A per-frequency-group decomposition localizes the gain precisely where the taxonomy was designed to act: Tail-group AP rises to 0.319, versus 0.269 for the control and 0.280 for the YOLOv11-l baseline, a control gap that a Tail-AP-specific paired test confirms is significant ($p_t = 0.006$; Section VI-F), while Head and Medium AP are unchanged. This is the intended CHARD mechanism, in which sharing supervision across a lower-cardinality attribute space lets rare classes that individually provide little gradient signal benefit from co-occurring attribute statistics, and it materializes once the backbone (via YOLOv11's C3k2 feature aggregation) is expressive enough to host the auxiliary representation without sacrificing the primary task. Because the attribute head is inference-free (Section VI-G), this tail-class gain carries zero deployment cost.

These findings refine, rather than contradict, the auxiliary-task literature. Attribute supervision is known to help when attributes cross-cut classes non-deterministically (color, pose, occlusion state) or enable open-vocabulary transfer [33], [36]. We add a second axis: even when the auxiliary labels are fully determined by the primary labels (the least favorable case, offering no orthogonal information), the auxiliary task can still act as a useful long-tail regularizer, provided the backbone has sufficient capacity to absorb it. The practical guidance for taxonomic auxiliary supervision is therefore to pair it with a high-capacity backbone and to target long-tailed regimes, exactly the setting in which CHARD-YOLOv11 delivers its gains. The cross-dataset results support this reading rather than undercutting it: the two supplementary datasets, on which characteristic supervision is neutral, contain no extreme tail for it to act upon (Section III-G): each covers only eight BadODD classes (omitting four of the five rarest (priority_vehicle, construction_vehicle, train, and wheelchair) as well as person, the most frequent class) and spans class frequencies within a single order of magnitude. The absence of an effect where there is no long tail is precisely what the mechanism predicts.

We state this operating condition carefully, because BadODD differs from the two supplementary datasets along more than one axis: it is both far more imbalanced (870:1 versus 26:1 and 2.3:1) and far denser (7.98 objects per image versus 2.56 and 1.00). With three datasets these two properties are confounded, and a cross-dataset comparison alone cannot attribute the effect to either. Two pieces of within-dataset evidence nevertheless favor class imbalance as the operative factor. First, the gain is not distributed across the label space but is localized to the rarest categories (Figure 7): wheelchair and construction_vehicle, two of the three scarcest classes in the dataset, account for essentially all of it, while Head and Medium classes move by at most 0.006. A scene-density mechanism would be expected to lift classes broadly rather than to concentrate on the least frequent ones. Second, a district-level analysis within BadODD finds no association between a district's object density and the size of the CHARD gain (Spearman $\rho = -0.18$, $p = 0.64$ across the nine districts), although the low power of that test at nine districts means it cannot exclude a density contribution. We therefore position the severity of class imbalance, meaning the extent to which the label space contains classes with very few instances, as the condition under which characteristic supervision pays off, and regard scene density as a plausible but unverified secondary factor that a larger and more diverse pool of datasets would be needed to separate.

For completeness, the end-to-end CHARD-RT-DETR variant underperforms substantially: it achieves only 0.5452 mAP_{50} on validation (Table 5), well below vanilla RT-DETR-l (0.7172). We emphasize that this comparison rests on a single training run of a custom pipeline built on a different backbone checkpoint than the RT-DETR-l baseline (Section IV-A), with no seed replication and under the fixed engineering budget noted in Section IX; we therefore treat it as a preliminary observation rather than a settled negative result. A plausible, though unverified, explanation is the difficulty of replacing a COCO-pretrained classification head with a freshly initialized hierarchical head while maintaining stable joint optimization; matched-query classification accuracy reached 99.8% during training, suggesting the bottleneck lies in localization rather than classification. RT-DETR's standard pipeline relies on denoising queries, multi-stage decoder supervision, and carefully tuned augmentation interactions that our custom variant could not fully replicate within the available engineering budget, which would be broadly consistent with the multi-seed finding (Section VI-F) that the RT-DETR family is more brittle than the YOLO detectors on this problem, though a single unreplicated run cannot establish this on its own.

C. GEOGRAPHIC DIFFICULTY VARIANCE AND DOMAIN SHIFT

The per-district analysis (Table S1, Supplementary Material) reveals a 2.22× performance gap between the easiest district (Mymensingh) and the hardest (the Maowa, Sirajganj, and Sherpur cluster). Averaged over three seeds, this gap is architecture-invariant: every model's easiest-to-hardest ratio falls between 2.02× and 2.12×, and all eight rank the five easiest districts identically. Inter-architecture variance within a district (0.03–0.05 mAP, up to 0.084) is far smaller than inter-district variance (0.29–0.65 mAP), so the choice of evaluation region affects the reported number far more than the choice of detector.

Plausible contributing factors include device-specific capture differences (resolution, stabilization, rolling shutter), traffic composition (dense, slow-moving traffic in Mymensingh and Dhaka versus small, motion-blurred vehicles on the Maowa expressway), and uneven tail-class distribution across districts (e.g., construction_vehicle is largely absent from Sirajganj and Sherpur). Disentangling their relative contributions is left to future work.

A single aggregate number can therefore mask large geographic variation: leading models' validation $\mathrm{mAP}_{50:95}$ of roughly 0.49 drops to 0.32–0.33 on the hardest single-district evaluations, about a third of the headline figure. This does not by itself establish that the standard random split leaks spatial priors between train and test splits; confirming that would require cross-district or leave-one-region-out training, which we recommend as a standard reporting axis for regional autonomous driving datasets and leave to future work.

D. TAIL-CLASS INSTABILITY

The Tail group exhibits a striking validation-to-test drop across all models. For YOLOv8-l, Tail AP falls from 0.422 on validation to 0.287 on test (−0.135), while Head and Medium AP change by less than 0.010. The same pattern holds for every other model (Table 6). This is not a model failure but a measurement artifact: the six Tail classes together account for only 423 test instances, and the three rarest (`construction_vehicle`, `train`, and `wheelchair`) contribute just 11, 16, and 28 instances respectively, with as few as 6, 5, and 27 on validation (Table 1). With so few support samples, per-class AP estimates have very high variance: a single additional detection on a class with six validation objects moves its AP by roughly 0.17, which is the same order as the entire spread between architectures. Consistent with this, the seed-to-seed standard deviation on these classes reaches 0.08–0.10, several times larger than the between-model gaps.

This volatility undermines the reliability of Tail metrics for cross-model comparison on BadODD specifically and on similarly imbalanced regional datasets in general. We recommend two mitigations for future work: first, reporting per-class instance counts alongside per-class AP so that readers can weight conclusions appropriately; second, augmenting evaluation with frequency-stratified protocols only when the Tail group has sufficient support, or using a separate few-shot detection metric (e.g., AP per instance encountered) for genuinely rare categories. Joint geographic-and-class balanced sampling during training may further stabilize learning on rare classes, but cannot rescue underpowered evaluation.

VIII. REPRODUCIBILITY AND RELEASED ARTIFACTS

Because several of the conclusions above rest on differences small enough to be confused with training noise, we consider the reproducibility of the study to be part of its contribution, and we describe here exactly what is fixed and what is released.

All random seeds (PyTorch, NumPy, Python) are fixed to 0 for the single-run tables of Section VI-A, and swept over {0, 1, 2} for the multi-seed study of Section VI-F. Training and evaluation configurations are recorded in YAML files alongside each model checkpoint. Per-epoch loss curves and validation metrics are saved to `results.csv` (Ultralytics models) and `log.json` (CHARD-RT-DETR); every seed-study run additionally logs val/test mAP, per-class AP, and controlled inference latency to a machine-readable benchmark file, so that every number in Tables 5–10 can be regenerated from the stored records rather than re-derived by hand.

We release the complete preprocessing pipeline (Section III), the dataset-conversion scripts that map Poribohon-BD and Sorokh-Poth onto the shared 13-class taxonomy (Section III-G), the class-to-attribute lookup that supplies CHARD's auxiliary targets (Section IV-B3), all training scripts, the evaluation scripts including per-district stratification and the multi-seed significance analysis, the figure-generation scripts, and the final checkpoints for every model reported in this paper.

Supplementary material accompanies this article, comprising the full per-district results table (Table S1) summarized in Section VI-B and discussed in Section VII-C. This material was included in the version submitted for peer review.

IX. LIMITATIONS

While our study establishes a rigorous benchmarking framework and exposes critical insights regarding multi-task optimization and spatial distribution shifts, several limitations must be acknowledged. First, although we validate our architectural conclusions across three independent Bangladeshi road-vehicle datasets (Section VI-F), all three originate from the same country and camera modality. The consistent architecture ranking we observe therefore demonstrates robustness to dataset and distribution shift within the Bangladeshi road ecosystem, but broader transfer to other regional South Asian benchmarks with different class taxonomies and road conventions, such as the Indian Driving Dataset (IDD), remains to be established. The per-district geographic difficulty pattern, in particular, is characterized here only for BadODD, which carries district metadata; replicating this stratified analysis on other benchmarks is left to future work. Relatedly, our claim that characteristic supervision pays off specifically in long-tailed regimes rests on a three-dataset comparison in which class imbalance and scene density covary, supplemented by within-BadODD per-class evidence (Section VII-B). Cleanly separating these two factors would require datasets that are long-tailed but sparse, or dense but balanced, which the currently available Bangladeshi benchmarks do not provide.

Second, our proposed CHARD framework, particularly the custom CHARD-RT-DETR variant, was constrained by a fixed computational and engineering budget. Due to these infrastructure limits, advanced transformer-specific training mechanics, such as denoising queries for stabilizing bipartite matching, were not integrated. Furthermore, standard complex data augmentations like Mosaic and MixUp failed to stabilize under our custom multi-task head configuration and were disabled for the RT-DETR variant. Consequently, the performance of CHARD-RT-DETR represents a baseline implementation rather than a fully optimized upper bound.

Finally, the study is bound by the nature of the data modality itself. The dataset relies entirely on 2D bounding box annotations extracted from smartphone cameras, which lack the 3D bounding boxes, depth estimation, and sensor fusion (e.g., LiDAR or Radar) required for full-scale autonomous vehicle deployment. Additionally, because the images are evaluated as isolated frames, our models do not exploit temporal or sequential context, which could otherwise mitigate localized occlusions and scale volatility through multi-frame tracking.

X. CONCLUSION

In this paper, we presented a comprehensive multi-protocol benchmark evaluating modern dense object detectors on the complex, unstructured driving environments of Bangladesh, hardened by a 72-model multi-seed study with formal statistical significance testing and cross-dataset replication on three independent datasets. Our empirical results demonstrate that YOLOv11-l stands among the co-leading baseline architectures (statistically tied with YOLOv8-l), with a statistically significant margin over YOLOv10-l and RT-DETR-l, and that all evaluated detectors meet real-time latency requirements. Our investigation into the CHARD framework reveals that the value of characteristic-aware supervision is architecture-dependent: on a capacity-limited backbone the deterministic taxonomy is redundant with the class labels and yields no gain, whereas on the stronger YOLOv11 backbone the same supervision acts as an effective long-tail regularizer, producing the best BadODD result of any evaluated model: a tail-class improvement (statistically significant against its no-attribute control) obtained at zero inference cost. Two conditions therefore govern whether a taxonomic auxiliary signal pays off: sufficient backbone capacity to absorb it, and a sufficiently long-tailed class distribution for it to act upon. On the two supplementary datasets, which are comparatively balanced, the same supervision is neutral, exactly as this account predicts.

Importantly, our per-district stratified evaluation revealed a profound $2\times$ performance gap between urban and high-speed expressway regions within the same country. This finding highlights the inadequacy of relying solely on aggregate, dataset-wide metrics, which inherently mask severe localized failure modes. We strongly argue that cross-district or cross-region evaluation must become a standard reporting axis for autonomous driving datasets to ensure robust spatial generalization. To foster reproducibility and encourage the community to build upon these geographic and structural insights, we have made our complete training pipelines, evaluation protocols, and pre-trained weights openly available.

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REFERENCES

[1] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? The KITTI vision benchmark suite," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Providence, RI, USA, Jun. 2012, pp. 3354–3361.

[2] M. Cordts, M. Omran, S. Ramos, T. Rehfeld, M. Enzweiler, R. Benenson, U. Franke, S. Roth, and B. Schiele, "The cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 3213–3223.

[3] F. Yu, H. Chen, X. Wang, W. Xian, Y. Chen, F. Liu, V. Madhavan, and T. Darrell, "BDD100K: A diverse driving dataset for heterogeneous multitask learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 2633–2642.

[4] P. Sun et al., "Scalability in perception for autonomous driving: Waymo open dataset," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2443–2451.

[5] G. Varma, A. Subramanian, A. Namboodiri, M. Chandraker, and C. V. Jawahar, "IDD: A dataset for exploring problems of autonomous navigation in unconstrained environments," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Waikoloa Village, HI, USA, Jan. 2019, pp. 1743–1751.

[6] M. N. Baig, R. Hajong, M. M. Patwary, M. S. Rahman, and H. A. Chowdhury, "BadODD: Bangladeshi autonomous driving object detection dataset," 2024, *arXiv:2401.10659*.

[7] G. Jocher, A. Chaurasia, and J. Qiu. (2023). *Ultralytics YOLOv8*. [Online]. Available: https://github.com/ultralytics/ultralytics

[8] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, J. Han, and G. Ding, "YOLOv10: Real-time end-to-end object detection," in *Proc. Adv. Neural Inf. Process. Syst. 37*, Vancouver, BC, Canada, Dec. 2024, pp. 107984–108011.

[9] R. Khanam and M. Hussain, "YOLOv11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.

[10] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen, "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2024, pp. 16965–16974.

[11] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft coco: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2014, pp. 740–755.

[12] R. Kumar, D. S. Reddy, and P. Rajalakshmi, "DriveIndia: An object detection dataset for diverse Indian traffic scenes," in *Proc. IEEE 28th Int. Conf. Intell. Transp. Syst. (ITSC)*, Nov. 2025, pp. 4249–4254.

[13] B. Saha, M. J. Islam, S. K. Mostaque, A. Bhowmik, T. K. Taton, M. N. H. Chowdhury, and M. B. I. Reaz, "Bangladeshi native vehicle detection in wild," 2024, *arXiv:2405.12150*.

[14] H. M. Hossain, P. Nath, M. N. Mahi, I. Uddin, I. J. Eiste, S. N. R. Ratul, M. N. U. Mozumdar, A. M. Saad, and M. T. Hossain, "Evaluating YOLO architectures: Implications for real-time vehicle detection in urban environments of Bangladesh," 2025, *arXiv:2509.05652*.

[15] J. Terven, D.-M. Córdova-Esparza, and J.-A. Romero-González, "A comprehensive review of YOLO architectures in computer vision: From YOLOv1 to YOLOv8 and YOLO-NAS," *Mach. Learn. Knowl. Extraction*, vol. 5, no. 4, pp. 1680–1716, Nov. 2023.

[16] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," 2025, *arXiv:2502.12524*.

[17] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. 16th Eur. Conf. Comput. Vis.*, vol. 12346. Cham, Switzerland: Springer, 2020, pp. 213–229.

[18] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," in *Proc. 9th Int. Conf. Learn. Represent. (ICLR)*, Virtual Event, Austria, 2021. [Online]. Available: arXiv:2010.04159

[19] H. Zhang, F. Li, S. Liu, L. Zhang, H. Su, J. Zhu, L. M. Ni, and H.-Y. Shum, "DINO: DETR with improved denoising anchor boxes for end-to-end object detection," in *Proc. 11th Int. Conf. Learn. Represent. (ICLR)*, Kigali, Rwanda, 2023. [Online]. Available: arXiv:2203.03605

[20] W. Lv, Y. Zhao, Q. Chang, K. Huang, G. Wang, and Y. Liu, "RT-DETRv2: Improved baseline with bag-of-freebies for real-time detection transformer," 2024, *arXiv:2407.17140*.

[21] Y. Peng, H. Li, P. Wu, Y. Zhang, X. Sun, and F. Wu, "D-FINE: Redefine regression task in DETRs as fine-grained distribution refinement," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Singapore, 2025. [Online]. Available: arXiv:2410.13842

[22] S. Huang, Z. Lu, X. Cun, Y. Yu, X. Zhou, and X. Shen, "DEIM: DETR with improved matching for fast convergence," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 15162–15171.

[23] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, "Class-balanced loss based on effective number of samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 9260–9269.

[24] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 2999–3007.

[25] J. Tan, C. Wang, B. Li, Q. Li, W. Ouyang, C. Yin, and J. Yan, "Equalization loss for long-tailed object recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 11659–11668.

[26] J. Wang, W. Zhang, Y. Zang, Y. Cao, J. Pang, T. Gong, K. Chen, Z. Liu, C. C. Loy, and D. Lin, "Seesaw loss for long-tailed instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9690–9699.

[27] A. Gupta, P. Dollár, and R. Girshick, "LVIS: A dataset for large vocabulary instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 5351–5359.

[28] Y. Zhang, B. Kang, B. Hooi, S. Yan, and J. Feng, "Deep long-tailed learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 10795–10816, Sep. 2023.

[29] Y. Ma, W. Dai, and J. Chen, "Pursuing better decision boundaries for long-tailed object detection via category information amount," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Singapore, 2025. [Online]. Available: arXiv:2502.03852

[30] P. V. Tran, "SimLTD: Simple supervised and semi-supervised long-tailed object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 4672–4681.

[31] K. P. Alexandridis, I. Elezi, J. Deng, A. Nguyen, and S. Luo, "Fractal calibration for long-tailed object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 15139–15150.

[32] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 6517–6525.

[33] L. Bertinetto, R. Müeller, K. Tertikas, S. Samangooei, and N. A. Lord, "Making better mistakes: Leveraging class hierarchies with deep networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 12503–12512.

[34] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 2980–2988.

[35] L. Liebel and M. Körner, "Auxiliary tasks in multi-task learning," 2018, *arXiv:1805.06334*.

[36] J. Deng, N. Ding, Y. Jia, A. Frome, K. Murphy, S. Bengio, Y. Li, H. Neven, and H. Adam, "Large-scale object classification using label relation graphs," in *Proc. 13th Eur. Conf. Comput. Vis.*, vol. 8689, Zurich, Switzerland. Cham, Switzerland: Springer, 2014, pp. 48–64.

[37] T. Standley, A. Zamir, D. Chen, L. J. Guibas, J. Malik, and S. Savarese, "Which tasks should be learned together in multi-task learning?" in *Proc. 37th Int. Conf. Mach. Learn.*, vol. 119, 2020, pp. 9120–9132.

[38] M. Kanakis, D. Brüggemann, S. Saha, S. Georgoulis, A. Obukhov, and L. V. Gool, "Reparameterizing convolutions for incremental multi-task learning without task interference," in *Proc. 16th Eur. Conf. Comput. Vis.*, vol. 12365. Cham, Switzerland: Springer, 2020, pp. 689–707.

[39] J. Jiang, B. Chen, J. Pan, X. Wang, D. Liu, J. Jiang, and M. Long, "ForkMerge: Mitigating negative transfer in auxiliary-task learning," in *Proc. Adv. Neural Inf. Process. Syst. 36*, 2023, pp. 30367–30389.

[40] S. Tabassum, S. Ullah, N. H. Al-Nur, and S. Shatabda, "Poribohon-BD: Bangladeshi local vehicle image dataset with annotation for classification," *Data Brief*, vol. 33, Dec. 2020, Art. no. 106465.

[41] T. Z. Sana, S. Islam, N. Hasan, and I. Ahmad, "Sorokh-poth: A balanced Bangladeshi road vehicle image dataset integrated with a detection system based on deep convolutional neural networks," *Int. J. Res. Publications*, vol. 122, no. 1, pp. 134–148, Apr. 2023.

[42] X. Li, W. Wang, L. Wu, S. Chen, X. Hu, J. Li, J. Tang, and J. Yang, "Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 21002–21012.

[43] C. Feng, Y. Zhong, Y. Gao, M. R. Scott, and W. Huang, "TOOD: Task-aligned one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Montreal, QC, Canada, Oct. 2021, pp. 3490–3499.

[44] H. W. Kuhn, "The Hungarian method for the assignment problem," *Nav. Res. Logistics Quart.*, vol. 2, nos. 1–2, pp. 83–97, 1955.

[45] H. Rezatofighi, N. Tsoi, J. Gwak, A. Sadeghian, I. Reid, and S. Savarese, "Generalized intersection over union: A metric and a loss for bounding box regression," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 658–666.

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